

API Testing Report

Postman-based End-to-End Testing

Project: E-Commerce REST API (.NET / ASP.NET Core)

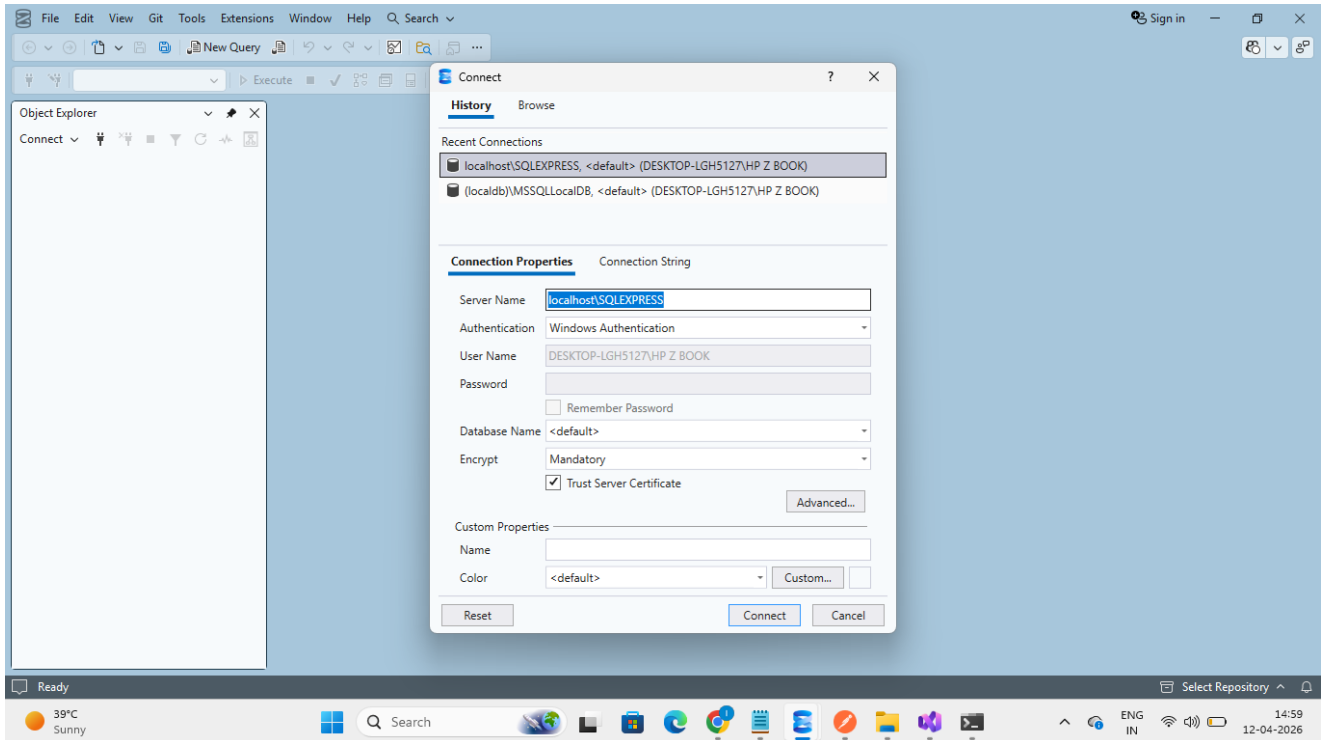
Tool Used: Postman

Section	Coverage
1. Setup & Prerequisites	SQL connection, project run checks
2. User Management	Admin & User Registration, Login
3. Product API	Create, Get All, Get by ID, Update, Delete
4. Order API	Place Order, Get All Orders, Get by ID
5. Advanced Features	Pagination, Search, Filtering
6. Error Reference	Common errors & troubleshooting guide

1. Setup & Prerequisites

1.1 SQL Server Connection

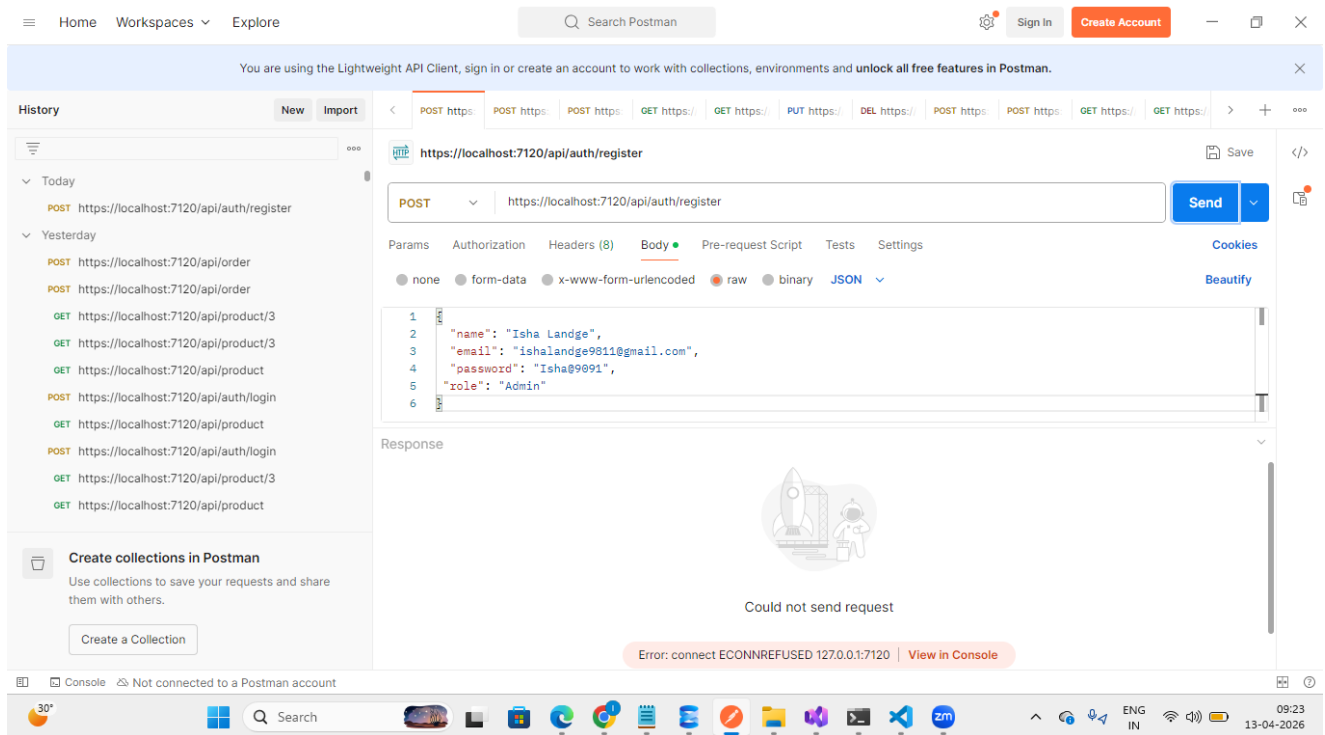
Before initiating any API tests, the SQL Server connection must be established. The connection string in the project configuration should reference the correct server name. Failure to establish this connection will result in all database-related endpoints being unavailable.



Screenshot 1.1 — SQL Server connection configured successfully.

1.2 Project Not Running — Common Pre-Test Error

A frequent mistake during testing is attempting to call API endpoints while the application is not running. If the project has not been started in Visual Studio (or the equivalent IDE), Postman will return a connection-refused error. Always ensure the project is running before executing any requests.



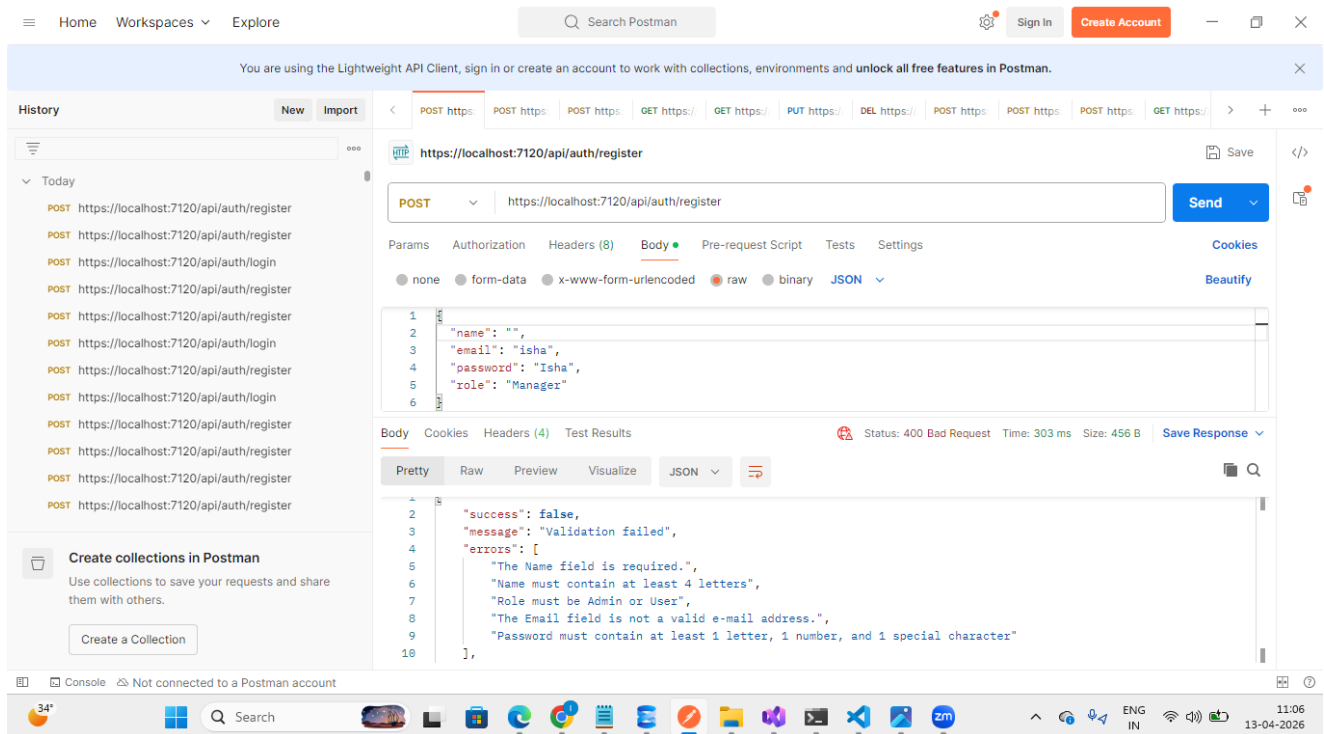
Screenshot 1.2 — Error observed when the project is not running and Postman tries to reach an endpoint.

2. User Management

2.1 Admin Registration

Validation Errors on Registration

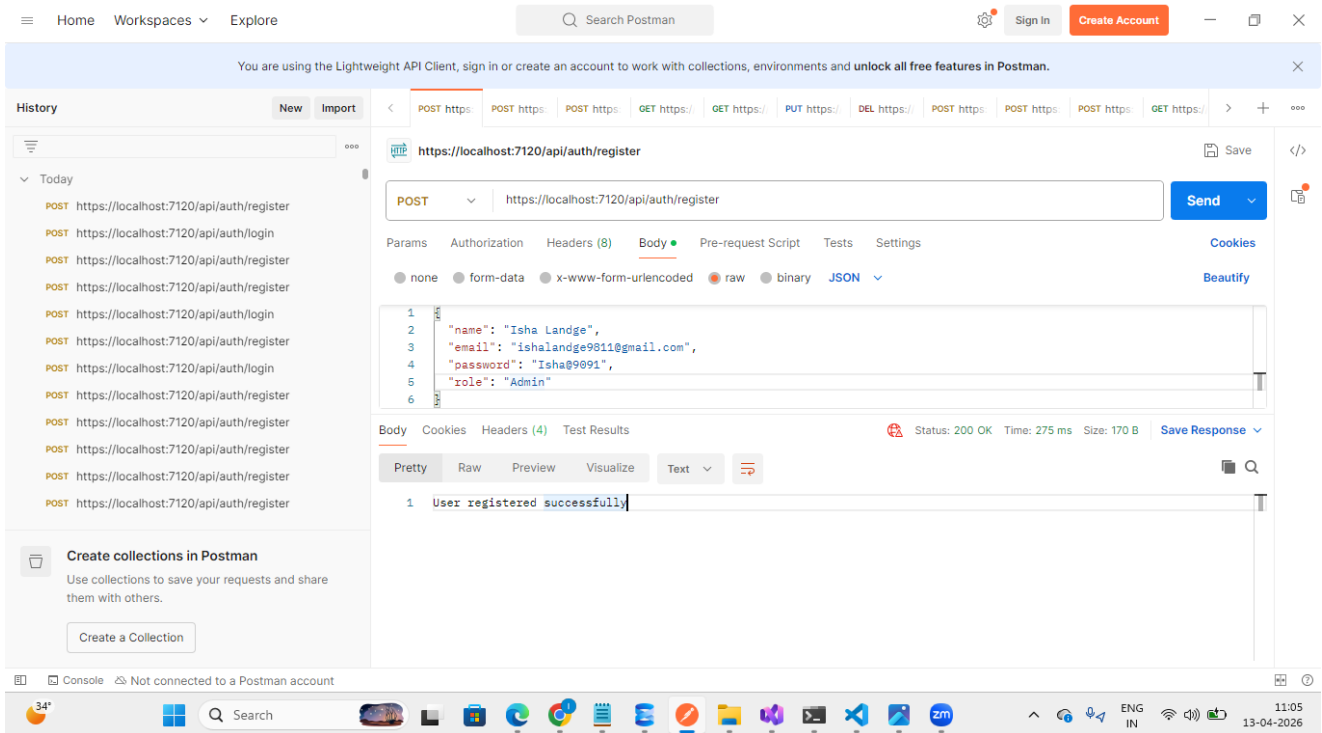
When required fields are missing or do not meet validation constraints (e.g., invalid email format, weak password, empty fields), the API returns a 400 Bad Request with descriptive validation messages to guide the client.



Screenshot 2.1a — Admin registration validation errors (400 Bad Request).

Successful Admin Registration

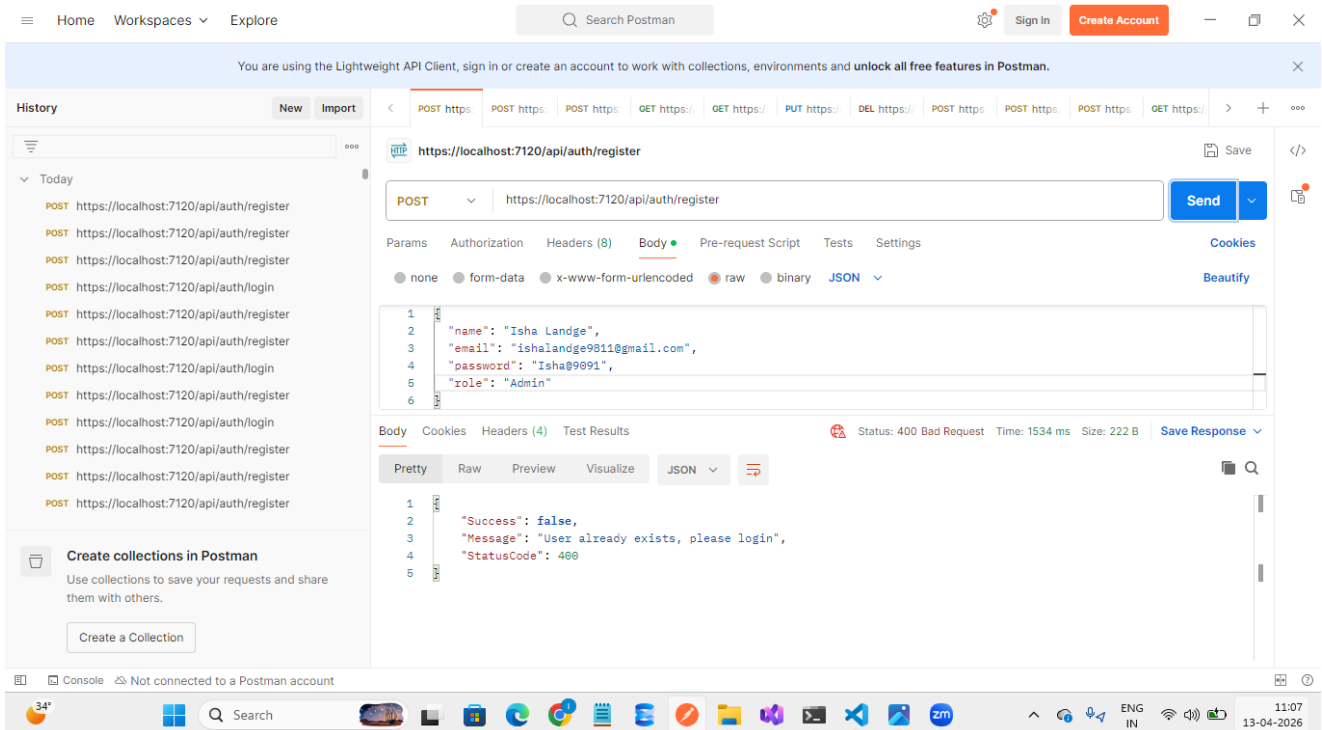
When all required fields are correctly provided, the admin account is created and the API returns a 200/201 success response confirming the registration.



Screenshot 2.1b — Admin user registered successfully.

Duplicate Registration Attempt

Attempting to register with an email address that already exists in the system returns a conflict error, preventing duplicate accounts.



Screenshot 2.1c — Error returned when registering an already-existing user.

Invalid Credentials

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⚙️

Sign In

Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

<POST https://POST https://POST https://GET https://GET https://PUT https://DEL https://POST https://POST https://POST https://GET https://> + ⋮

▼ Today

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/register

🗑️ Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

🔗 https://localhost:7120/api/auth/login

POST

https://localhost:7120/api/auth/login

Send

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

● none

● form-data

● x-www-form-urlencoded

● raw

● binary

JSON

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Successful Admin Login

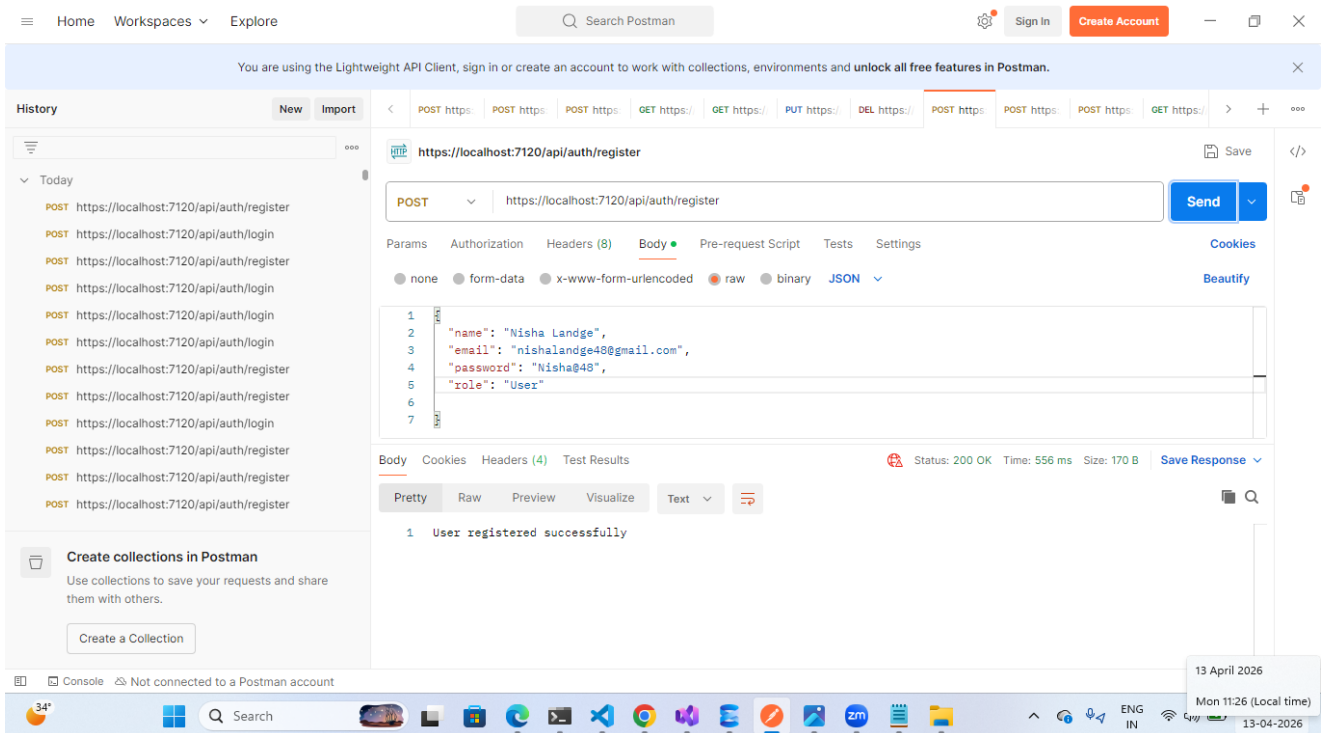
[illegible]

Screenshot 2.2b — Admin login successful; JWT token returned.

2.3 User Registration & Login

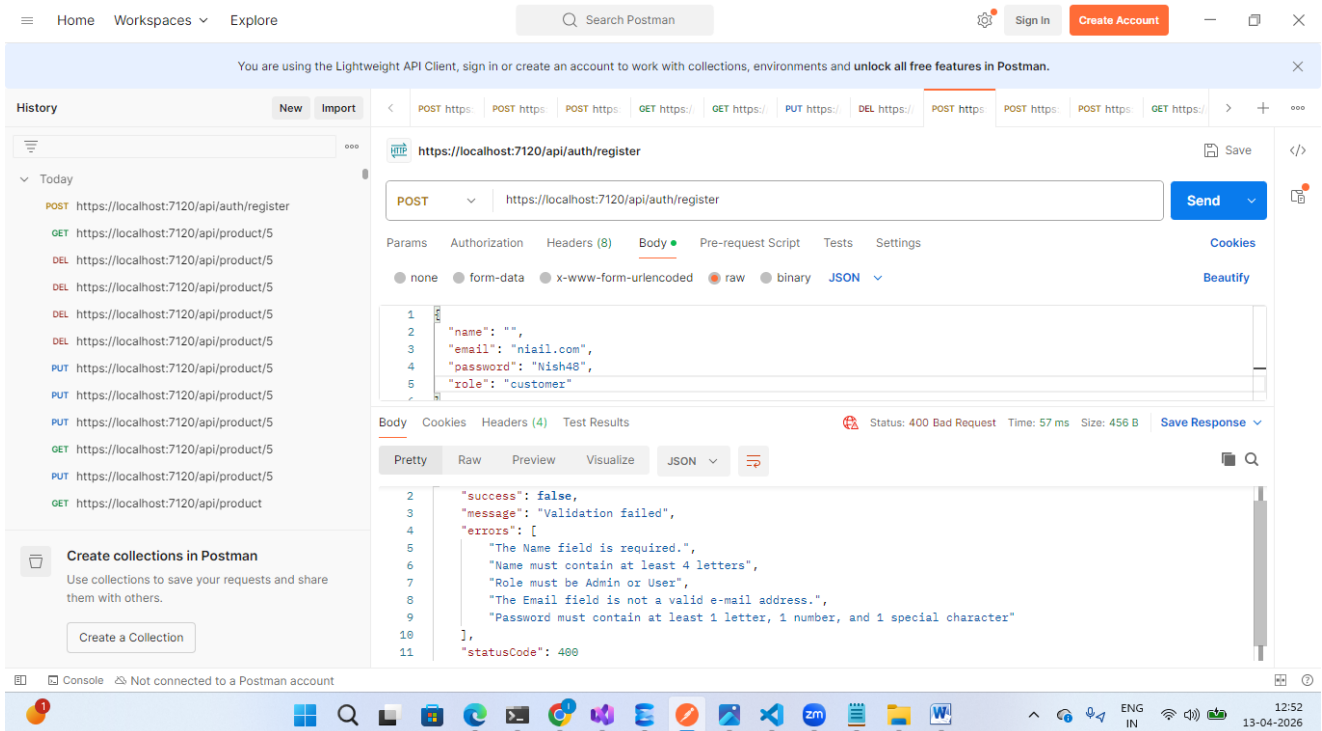
Successful User Registration

Regular user accounts can be created through the user registration endpoint. Upon success, the response confirms the account creation.



Screenshot 2.3a — Regular user registered successfully.

User Registration Validation Error



Screenshot 2.3b — Validation error during user registration (missing or invalid fields).

User Login — Successful (JWT Issued)

[illegible]

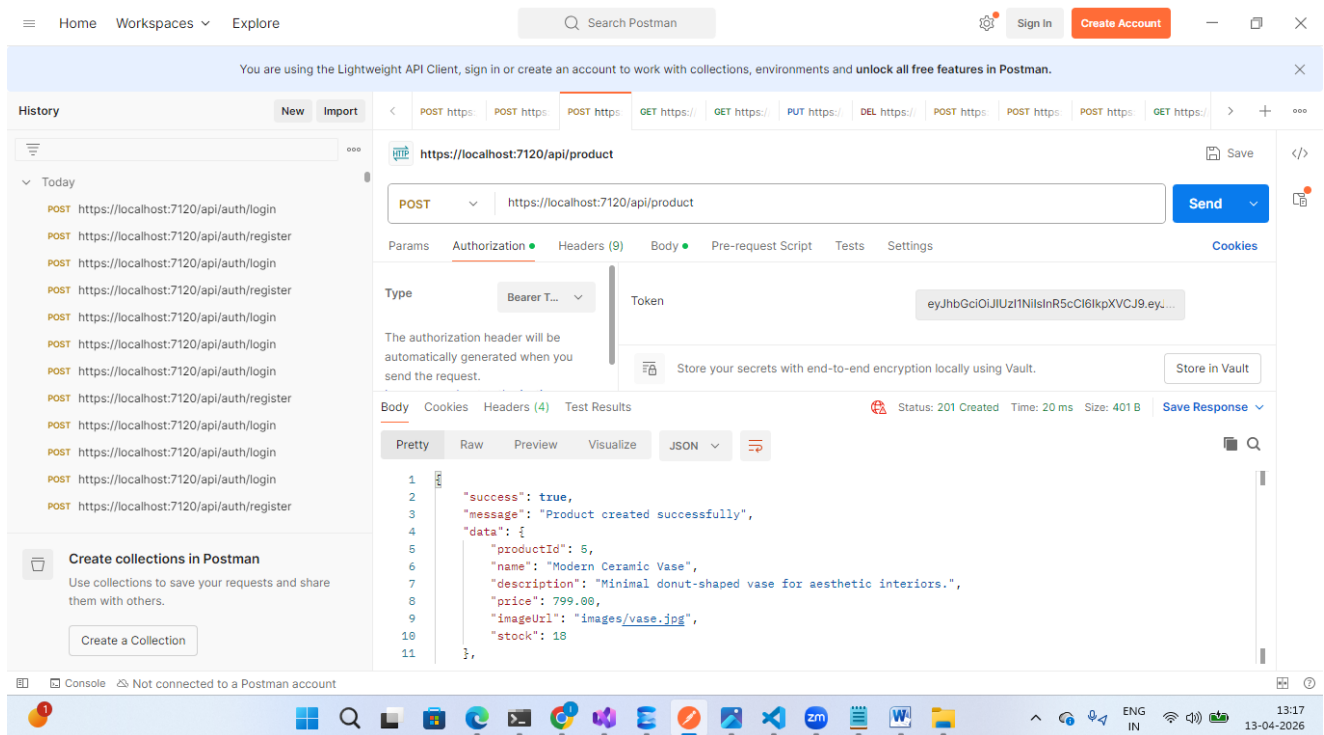
Screenshot 2.3d — User login successful; JWT token issued for subsequent requests.

3. Product API

3.1 Create Product (Admin Only)

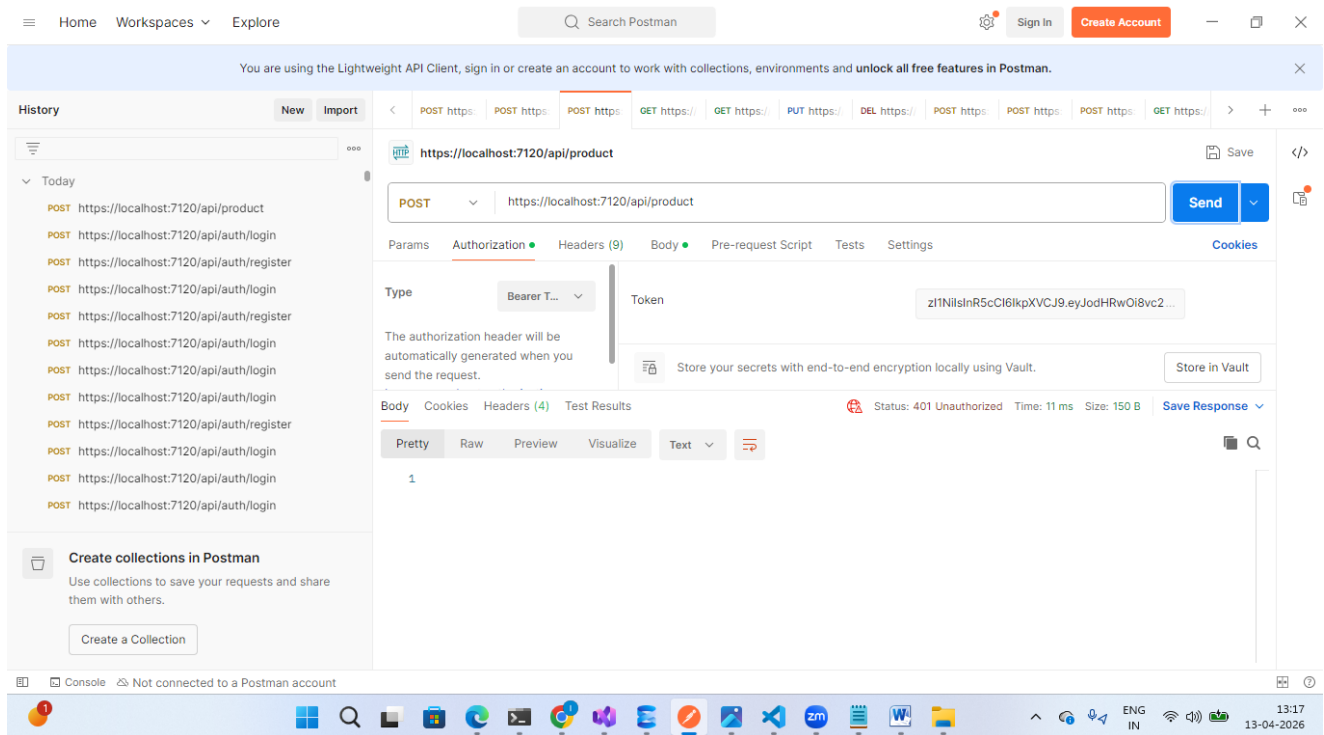
Product creation is restricted to administrators. The admin JWT token must be included in the Authorization header. The following test cases cover the three possible authorization states.

Correct Admin JWT Token



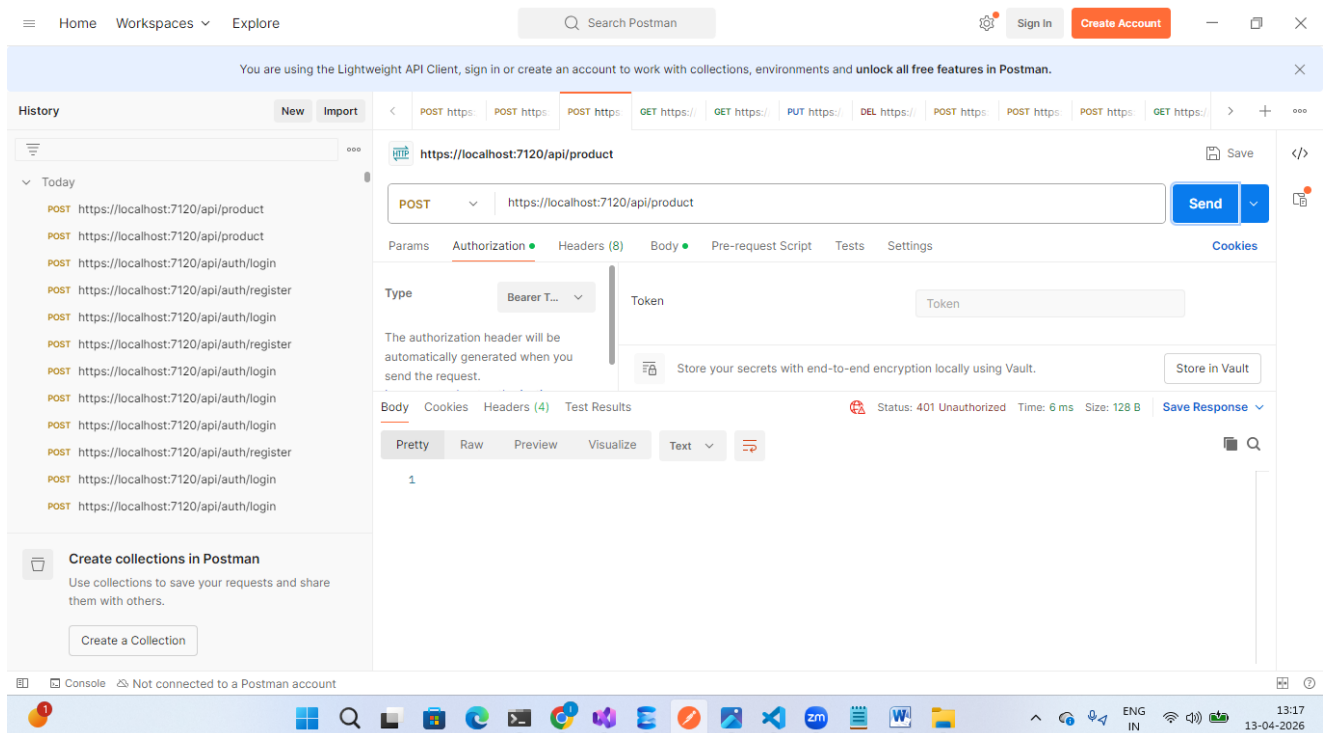
Screenshot 3.1a — Valid admin JWT token added to Authorization header.

Invalid JWT Token



Screenshot 3.1b — Invalid or tampered JWT token returns 401 Unauthorized.

No Authorization Token



Screenshot 3.1c — Missing Authorization header returns 401 Unauthorized.

Validation Failed on Product Creation

If required product fields (name, price, stock, etc.) are missing or contain invalid values, the API rejects the request with validation error details.

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Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

<POST https:POST https:POST https:GET https:GET https:PUT https:DEL https:POST https:POST https:POST https:GET https:>+

Today

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/auth/register

POST https://localhost:7120/api/product

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/product

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/register

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

https://localhost:7120/api/product

Save</>

POSThttps://localhost:7120/api/product

Send

ParamsAuthorizationHeaders (9)BodyPre-request ScriptTestsSettings

noneform-datax-www-form-urlencodedrawbinaryJSON

1234567

BodyCookiesHeaders (4)Test Results

Status: 400 Bad RequestTime: 20 msSize: 336 BSave Response

PrettyRawPreviewVisualizeJSON

123456789

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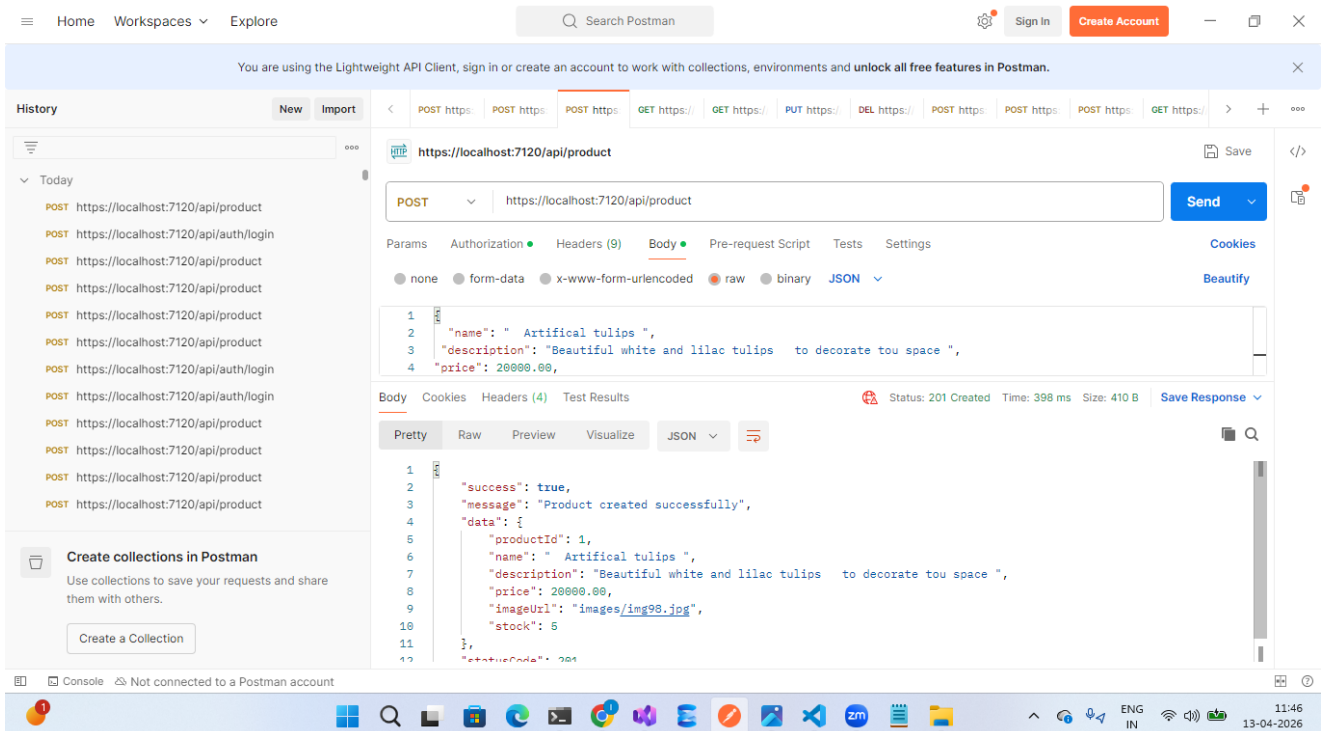
Windows Taskbar Icons

11:39 13-04-2026

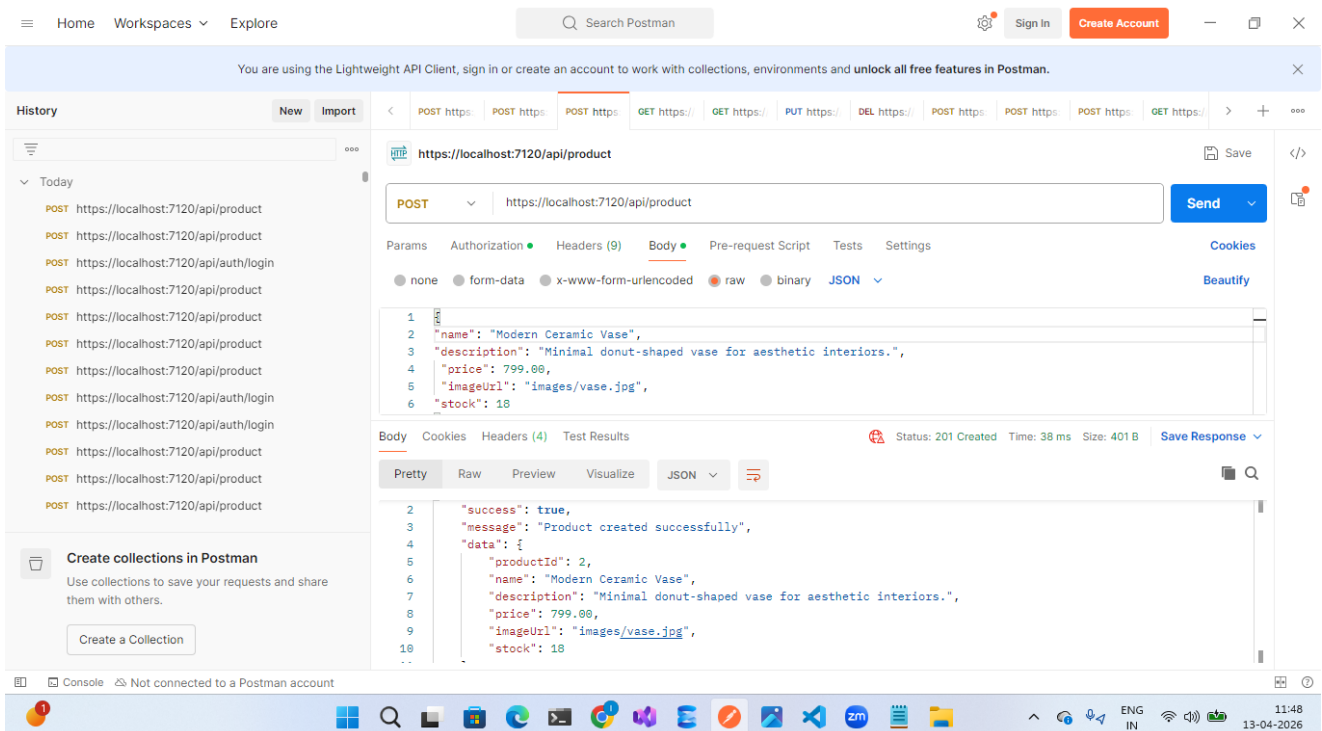
Screenshot 3.1d — Product creation failed due to validation errors.

Products Created (Product 1–5)

Five products were created during this test run. Products 3, 4, and 5 intentionally share the same name ('Modern Ceramic Vase') to verify that update and delete operations target the correct record by ID.



Screenshot 3.1e — Product 1 (Artificial Tulips) created successfully.



Screenshot 3.1f — Product 2 (Modern Ceramic Vase) created successfully.

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You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History: POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/auth/login, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/auth/login, POST https://localhost:7120/api/auth/login, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product

Create collections in Postman: Use collections to save your requests and share them with others. Create a Collection

Request: POST https://localhost:7120/api/product

Body: {"name": "Premium Sofa Set", "description": "Comfortable fabric sofa set with wooden frame and modern design.", "price": 32000.00, "imageUrl": "images/sofa.jpg", "stock": 3}

Response: {"success": true, "message": "Product created successfully", "data": {"productId": 3, "name": "Premium Sofa Set", "description": "Comfortable fabric sofa set with wooden frame and modern design.", "price": 32000.00, "imageUrl": "images/sofa.jpg", "stock": 3}}

Status: 201 Created Time: 26 ms Size: 413 B Save Response

Console: Not connected to a Postman account

Screenshot 3.1g — Product 3 (Premium Sofa Set) created successfully.

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You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History: POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/auth/login, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product, POST https://localhost:7120/api/auth/login, POST https://localhost:7120/api/auth/login, POST https://localhost:7120/api/product, POST https://localhost:7120/api/product

Create collections in Postman: Use collections to save your requests and share them with others. Create a Collection

Request: POST https://localhost:7120/api/product

Body: {"name": "Modern Ceramic Vase", "description": "Minimal donut-shaped vase for aesthetic interiors.", "price": 799.00, "imageUrl": "images/vase.jpg", "stock": 18}

Response: {"success": true, "message": "Product created successfully", "data": {"productId": 4, "name": "Modern Ceramic Vase", "description": "Minimal donut-shaped vase for aesthetic interiors.", "price": 799.00, "imageUrl": "images/vase.jpg", "stock": 18}}

Status: 201 Created Time: 32 ms Size: 401 B Save Response

Console: Not connected to a Postman account

Screenshot 3.1h — Product 4 (Modern Ceramic Vase) created successfully.

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You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

Today

- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/register
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/login

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

https://localhost:7120/api/product

GET https://localhost:7120/api/product

Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Type Bearer Token

The authorization header will be automatically generated when you send the request.

Token eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ...

Store your secrets with end-to-end encryption locally using Vault. Store in Vault

Body Cookies Headers (4) Test Results

Status: 200 OK Time: 60 ms Size: 1.09 KB Save Response

Pretty Raw Preview Visualize JSON

```
36  },
37  {
38    "productId": 5,
39    "name": "Modern Ceramic Vase",
40    "description": "Minimal donut-shaped vase for aesthetic interiors.",
41    "price": 799.00,
42    "imageUrl": "images/vase.jpg",
43    "stock": 18
44  },
45  ],
46  "statusCode": 200
```

Console Not connected to a Postman account

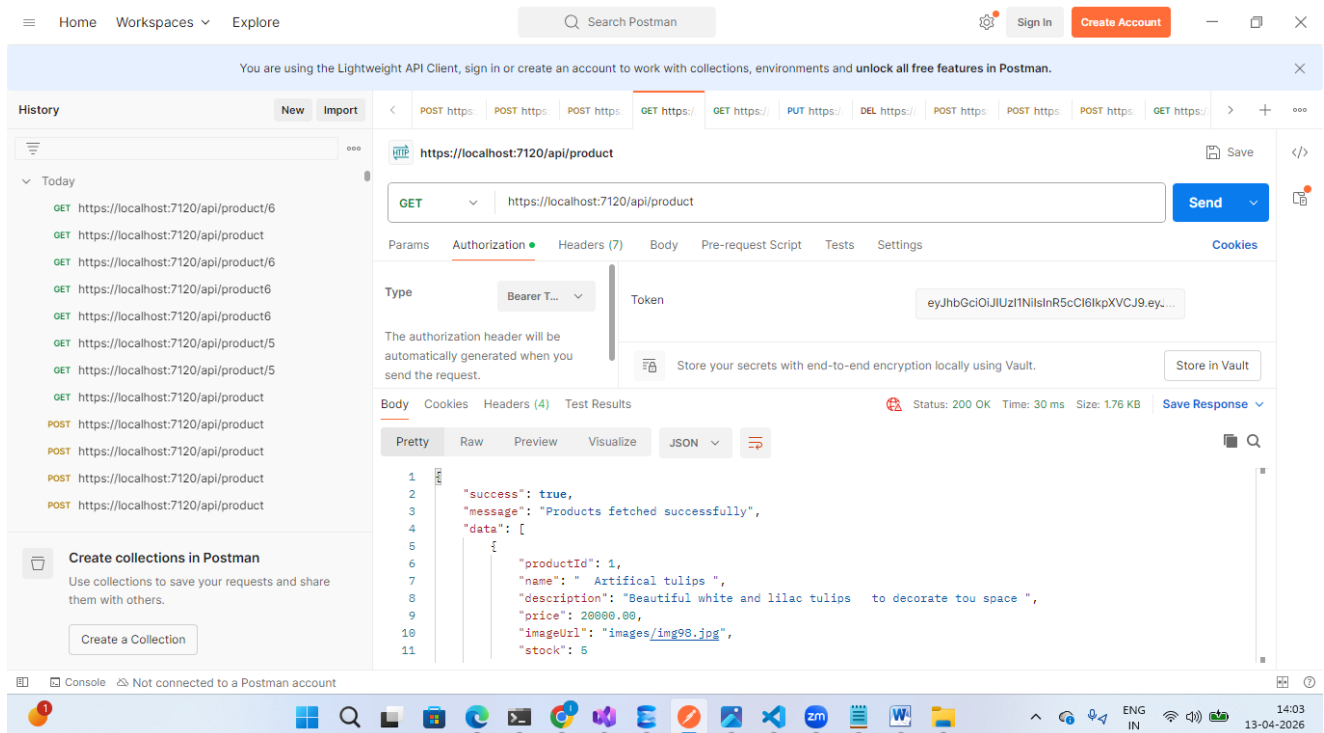
13:43 13-04-2026

Screenshot 3.1i — Product 5 (Modern Ceramic Vase) created successfully.

3.2 Get All Products

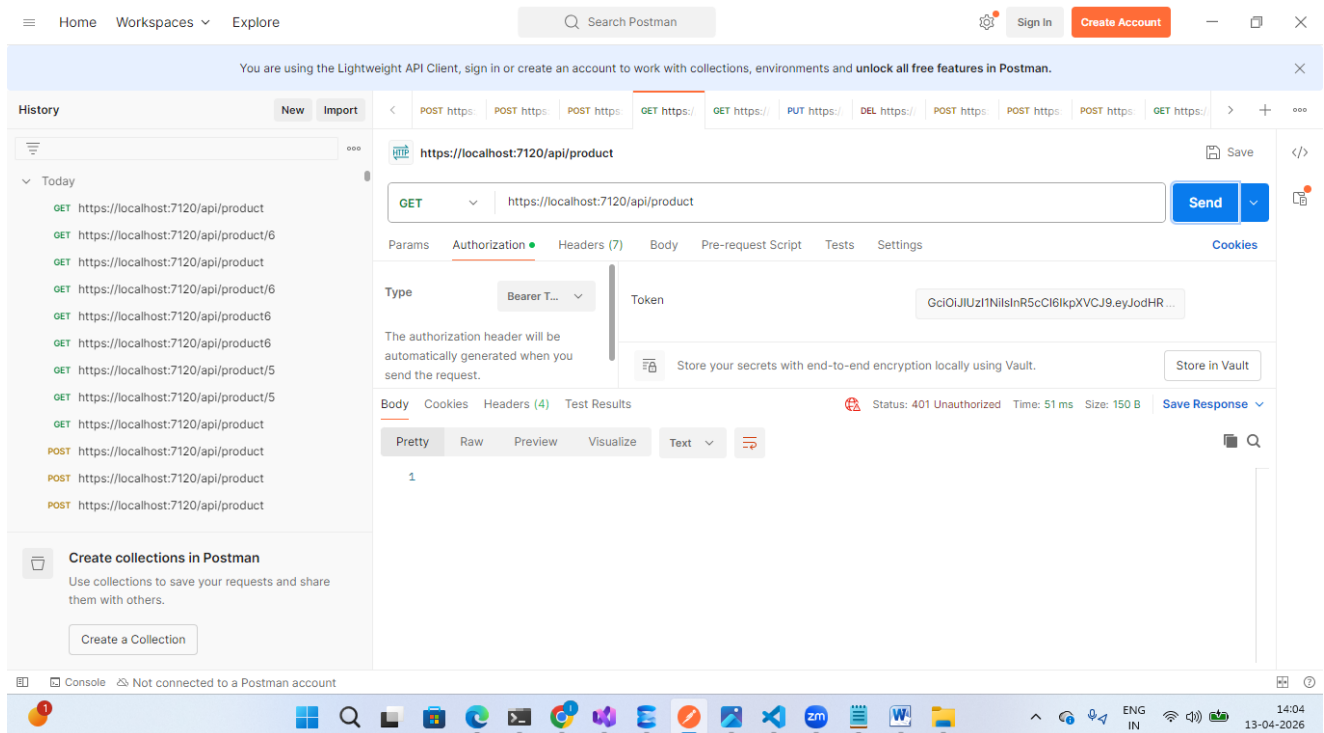
Both admin and user JWT tokens are accepted for retrieving all products. The endpoint is also accessible without authorization (configured with `[AllowAnonymous]` — see Section 5).

Correct JWT Token



Screenshot 3.2a — All products fetched with valid JWT token (200 OK).

Incorrect JWT Token



Screenshot 3.2b — Incorrect JWT token returns 401 Unauthorized.

Missing JWT Token

This screenshot shows the Postman interface with a GET request to `https://localhost:7120/api/product`. The Authorization tab is selected, but no token is entered. The response status is 401 Unauthorized, indicating that the request failed due to missing authentication.

History: Today

- GET https://localhost:7120/api/product
- GET https://localhost:7120/api/product
- GET https://localhost:7120/api/product/6
- GET https://localhost:7120/api/product
- GET https://localhost:7120/api/product/6
- GET https://localhost:7120/api/product/6
- GET https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product

Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection

GET https://localhost:7120/api/product

Params Authorization Headers (6) Body Pre-request Script Tests Settings Cookies

Type Bearer T... Token

The authorization header will be automatically generated when you send the request.

Store your secrets with end-to-end encryption locally using Vault. Store in Vault

Body Cookies Headers (4) Test Results Status: 401 Unauthorized Time: 2 ms Size: 128 B Save Response

Pretty Raw Preview Visualize Text

1

Console Not connected to a Postman account

14:04 13-04-2026

Screenshot 3.2c — No Authorization header returns 401 Unauthorized.

Products Fetched Successfully

This screenshot shows the Postman interface with the same GET request to `https://localhost:7120/api/product`. This time, a valid JWT token is entered in the Authorization tab. The response status is 200 OK, and the response body contains a JSON array of product data.

History: Today

- GET https://localhost:7120/api/product
- POST https://localhost:7120/api/auth/login
- GET https://localhost:7120/api/product
- GET https://localhost:7120/api/product
- GET https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product

Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection

GET https://localhost:7120/api/product

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Type Bearer T... Token eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1b250aW50IjpbImF1dG8iXSwiaWF0IjoxNzE0MzU0MDA0LCJpc2MiOiJ1b250aW50In0.eyJ1b250aW50IjpbImF1dG8iXSwiaWF0IjoxNzE0MzU0MDA0LCJpc2MiOiJ1b250aW50In0.

Body Cookies Headers (4) Test Results Status: 200 OK Time: 52 ms Size: 1.06 KB Save Response

Pretty Raw Preview Visualize JSON

```
1
2  {
3    "success": true,
4    "message": "Products fetched successfully",
5    "data": [
6      {
7        "productId": 1,
8        "name": "Artificial tulips ",
9        "description": "Beautiful white and lilac tulips  to decorate tou space ",
10       "price": 20000.00,
11       "imageUrl": "images/img98.jpg",
12       "stock": 5
13     },
14     {
15       "productId": 2,
16       "name": "Modern Ceramic Vase".
```

Console Not connected to a Postman account

12:02 13-04-2026

Screenshot 3.2d — Get all products response (page 1).

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You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History: Today
GET https://localhost:7120/api/product
POST https://localhost:7120/api/auth/login
GET https://localhost:7120/api/product
GET https://localhost:7120/api/product
GET https://localhost:7120/api/product
POST https://localhost:7120/api/product
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POST https://localhost:7120/api/product

Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection

https://localhost:7120/api/product
GET https://localhost:7120/api/product
Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Type: Bearer Token
Token: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ...

Body: Pretty Raw Preview Visualize JSON
Status: 200 OK Time: 52 ms Size: 1.06 KB Save Response

```
14 {
15   "productId": 2,
16   "name": "Modern Ceramic Vase",
17   "description": "Minimal donut-shaped vase for aesthetic interiors.",
18   "price": 799.00,
19   "imageUrl": "images/vase.jpg",
20   "stock": 18
21 },
22 {
23   "productId": 3,
24   "name": "Premium Sofa Set",
25   "description": "Comfortable fabric sofa set with wooden frame and modern design.",
26   "price": 32000.00,
27   "imageUrl": "images/sofa.jpg",
28   "stock": 3
29 }
```

Screenshot 3.2e — Get all products response (page 2).

Home Workspaces Explore Search Postman

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History: Today
GET https://localhost:7120/api/product
POST https://localhost:7120/api/auth/login
GET https://localhost:7120/api/product
GET https://localhost:7120/api/product
GET https://localhost:7120/api/product
POST https://localhost:7120/api/product
POST https://localhost:7120/api/product
POST https://localhost:7120/api/product
POST https://localhost:7120/api/product
POST https://localhost:7120/api/product

Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection

https://localhost:7120/api/product
GET https://localhost:7120/api/product
Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

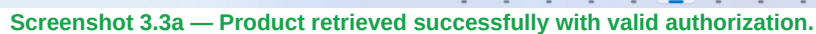
Type: Bearer Token
Token: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ...

Body: Pretty Raw Preview Visualize JSON
Status: 200 OK Time: 52 ms Size: 1.06 KB Save Response

```
29 {
30   "productId": 4,
31   "name": "Modern Ceramic Vase",
32   "description": "Minimal donut-shaped vase for aesthetic interiors.",
33   "price": 799.00,
34   "imageUrl": "images/vase.jpg",
35   "stock": 18
36 },
37 {
38   "productId": 5,
39   "name": "Modern Ceramic Vase",
40   "description": "Minimal donut-shaped vase for aesthetic interiors.",
41   "price": 799.00,
42   "imageUrl": "images/vase.jpg",
43   "stock": 18
44 }
```

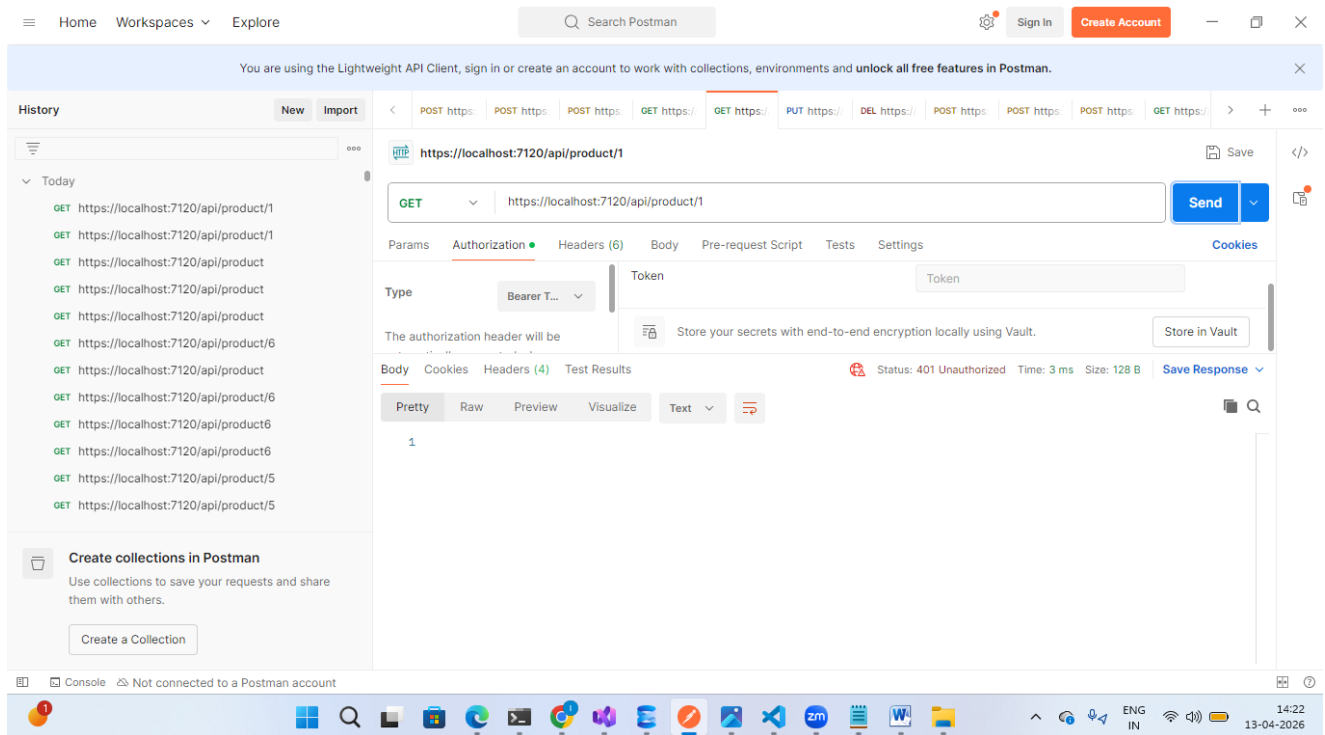
Screenshot 3.2f — Get all products response (page 3).

Correct Authorization



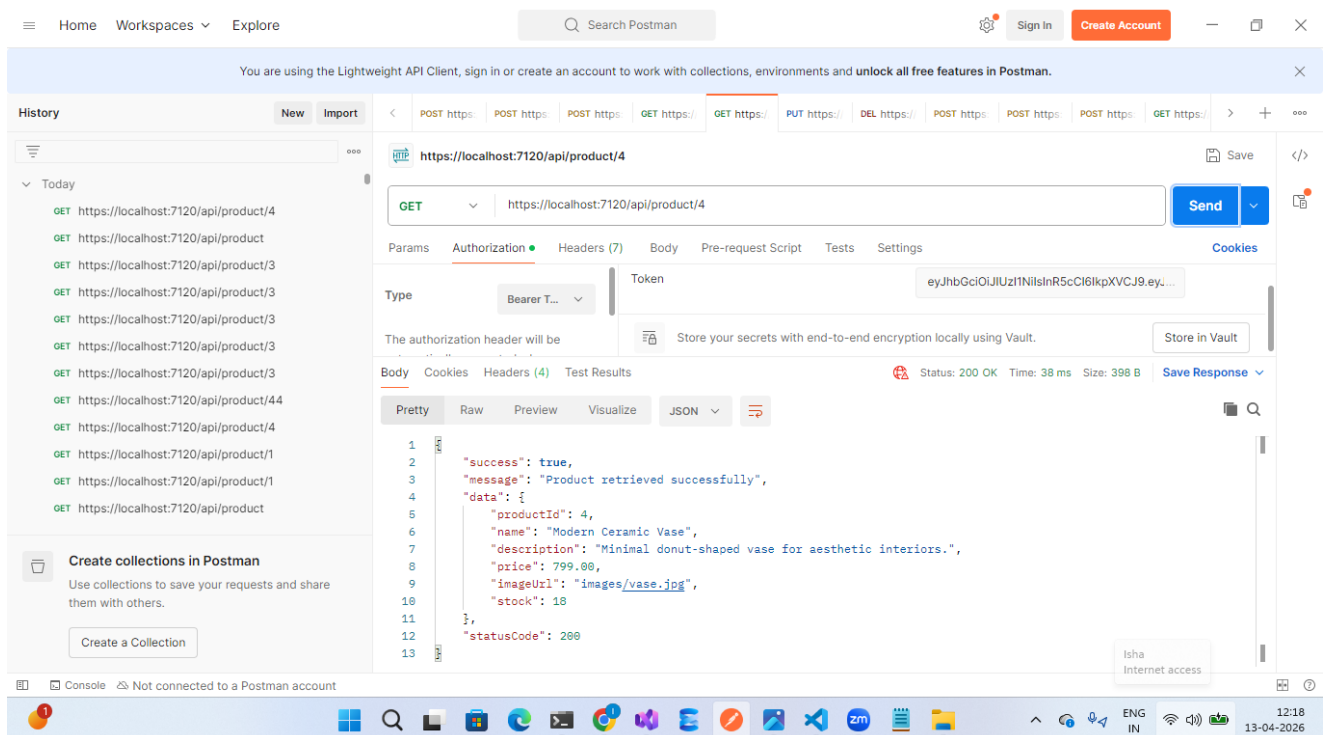
Screenshot 3.3b — Incorrect JWT token returns 401 Unauthorized.

Missing Authorization

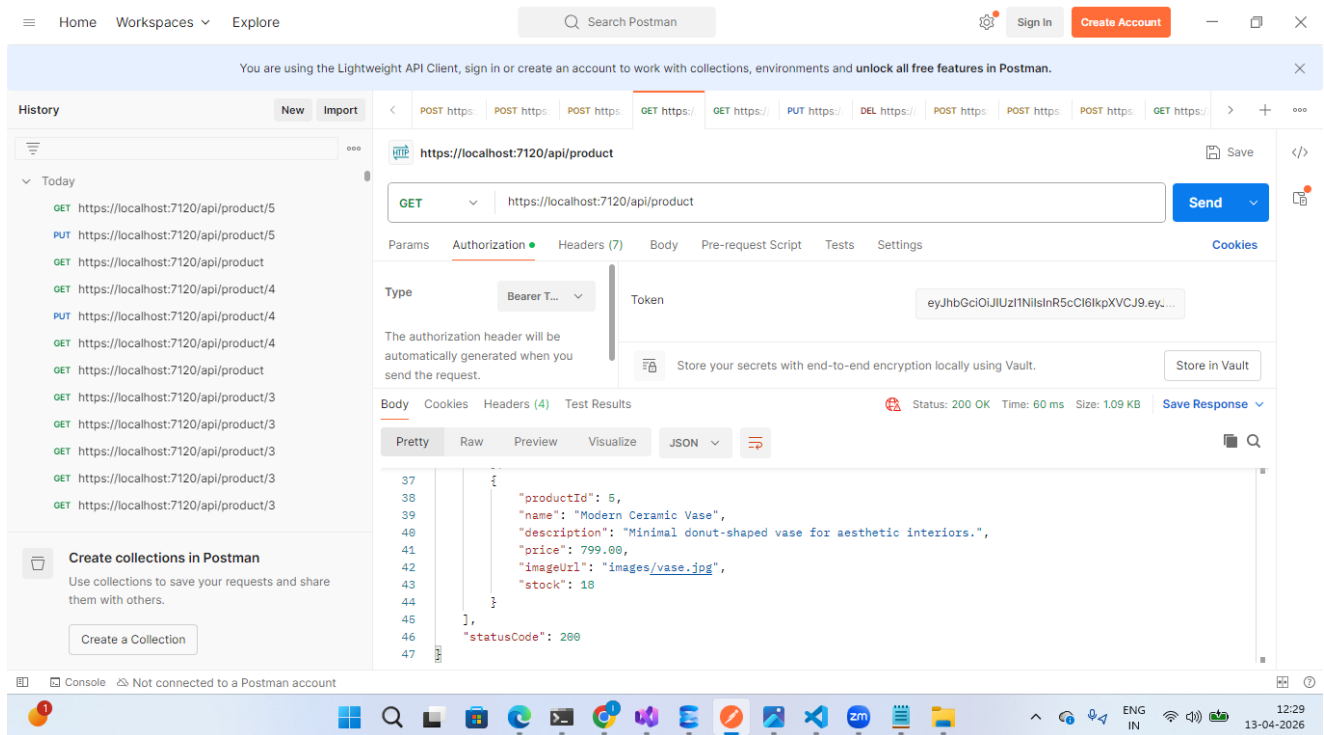


Screenshot 3.3c — Missing Authorization header returns 401 Unauthorized.

Get Product ID 4 & 5 (Before Update)



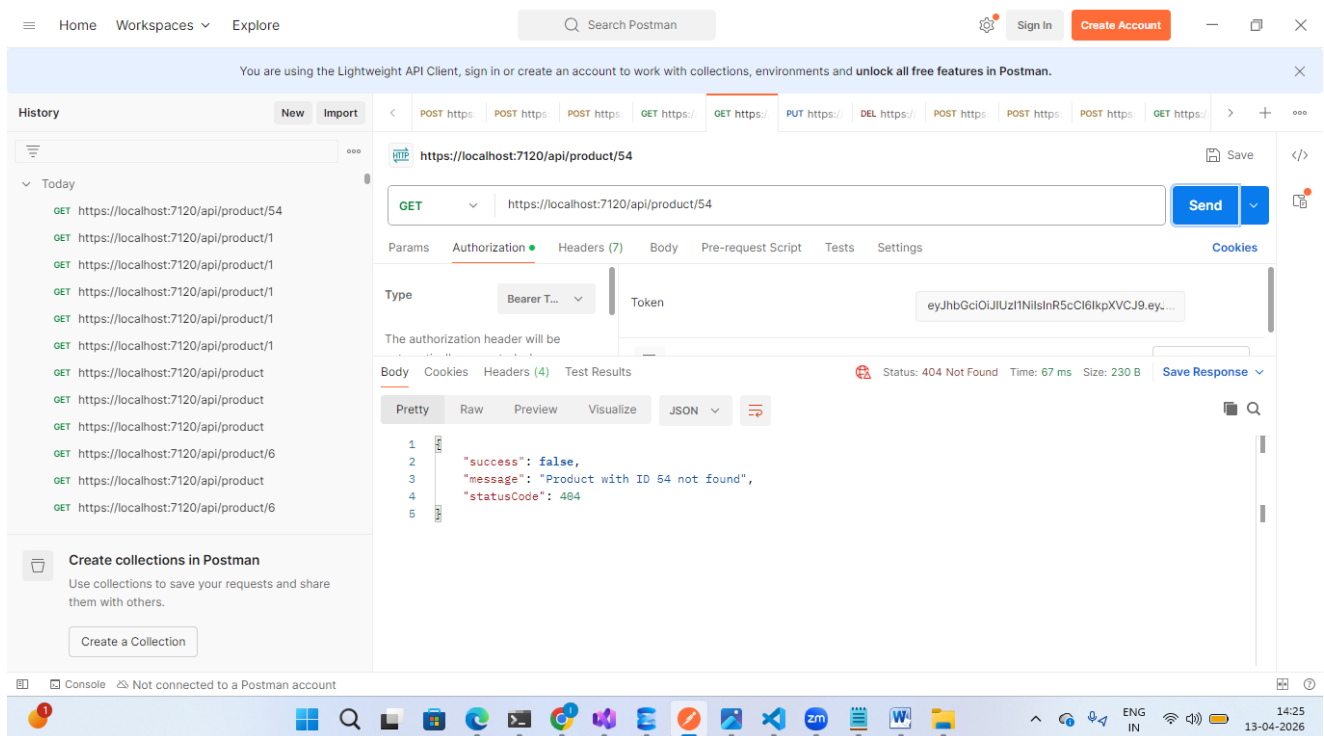
Screenshot 3.3d — Product ID 4 details before update.



Screenshot 3.3e — Product ID 5 details before update.

Product Not Found (ID 54)

Requesting a product ID that does not exist returns a 404 Not Found response.

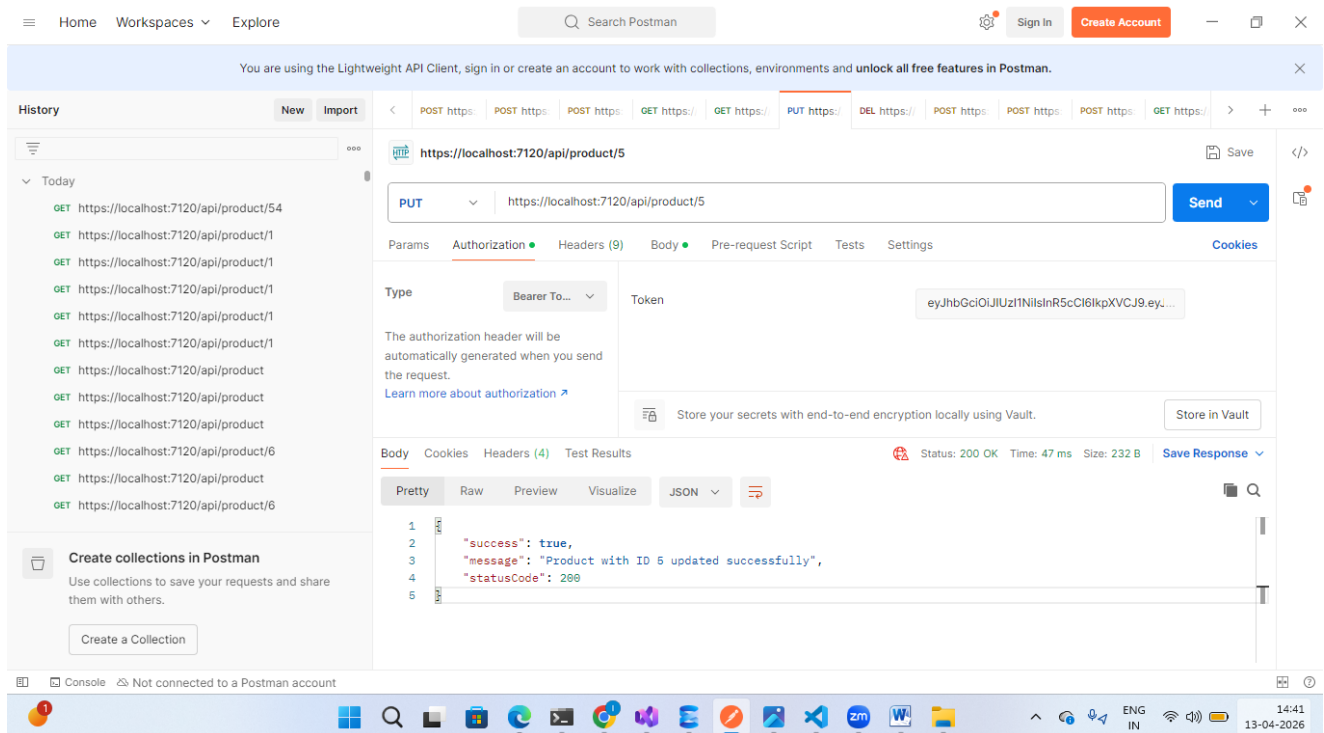


Screenshot 3.3f — 404 Not Found for non-existent product ID 54.

3.4 Update Product by ID (Admin Only)

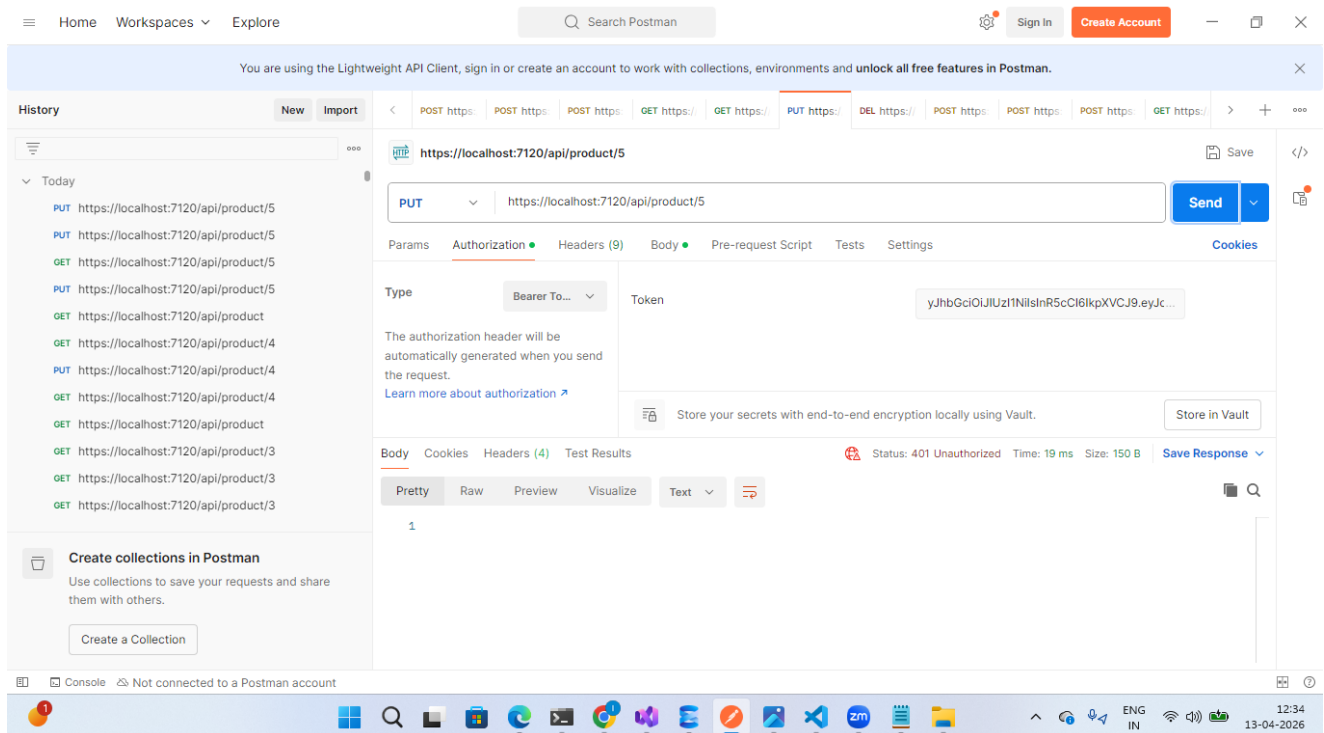
Only admin tokens are accepted for product updates. User tokens or missing tokens will be rejected with a 401/403 response.

Correct Admin Token



Screenshot 3.4a — Valid admin JWT token used for update request.

Wrong / User JWT Token



Screenshot 3.4b — User token or wrong token results in 403 Forbidden.

Missing Authorization

The screenshot shows the Postman interface with a PUT request to `https://localhost:7120/api/product/5`. The request is configured with the 'Authorization' tab selected, showing 'Bearer To...' and a 'Token' field. The response status is 401 Unauthorized, with a message indicating that the token is missing. The response body is empty. The status bar at the bottom shows 'Status: 401 Unauthorized', 'Time: 28 ms', and 'Size: 128 B'.

Screenshot 3.4c — Missing token returns 401 Unauthorized.

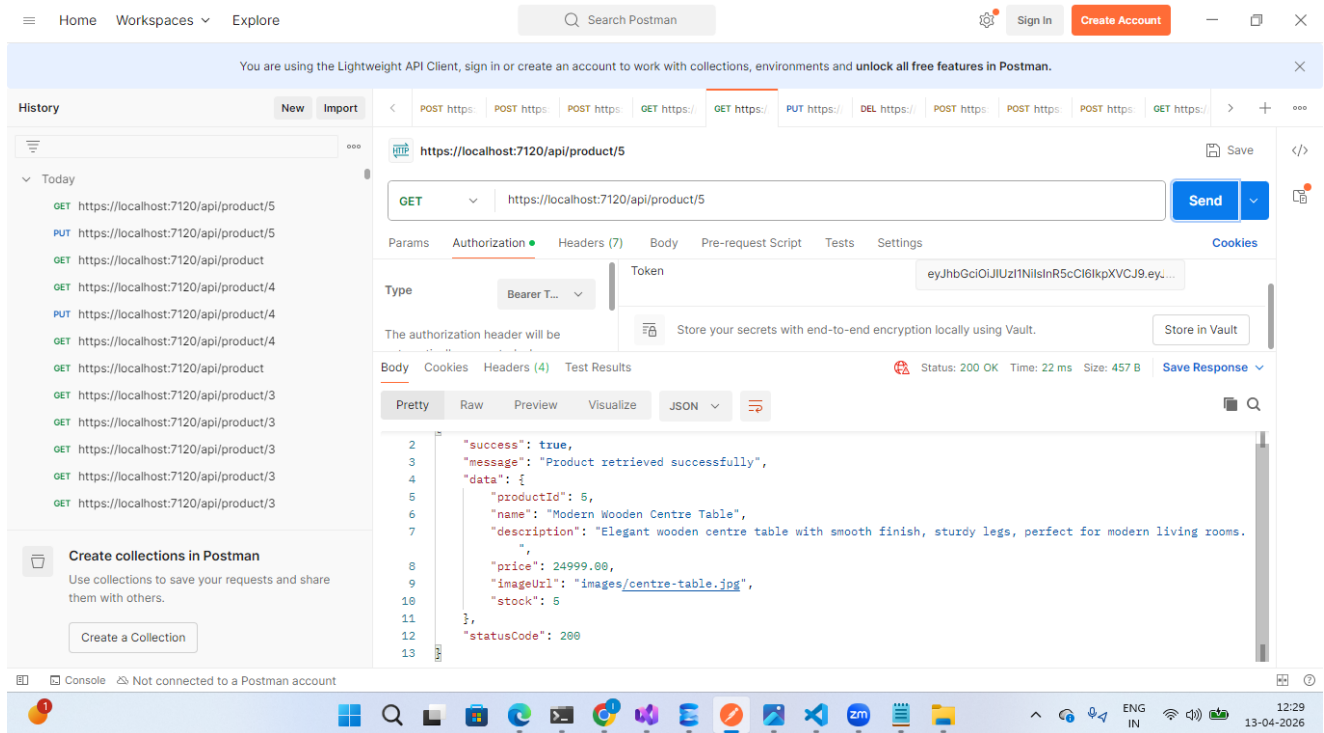
Update Response — Product ID 4

The update endpoint does not return the updated product in the response body. It only confirms that the update was successful. To verify the change, use the Get Product by ID endpoint.

The screenshot shows the Postman interface with a PUT request to `https://localhost:7120/api/product/4`. The request is configured with the 'Body' tab selected, showing a JSON payload for an indoor monstera plant. The response status is 200 OK, with a message indicating that the product was updated successfully. The response body contains a success message and the status code. The status bar at the bottom shows 'Status: 200 OK', 'Time: 17 ms', and 'Size: 232 B'.

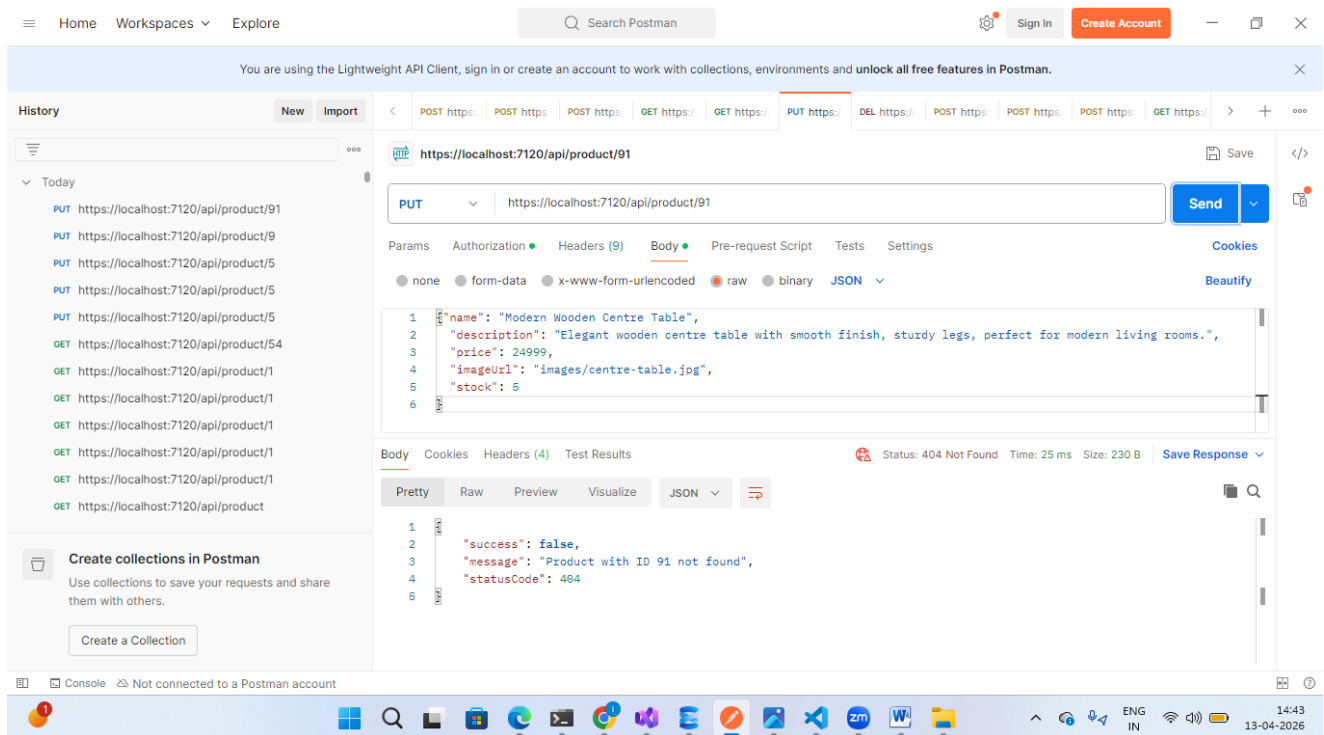
Screenshot 3.4d — Update confirmation message for product ID 4.

Verification — Updated Product ID 4



Screenshot 3.4g — Product ID 5 now shows updated data (Ceramic Vase → Modern Wooden Centre Table).

Update on Non-Existent Product

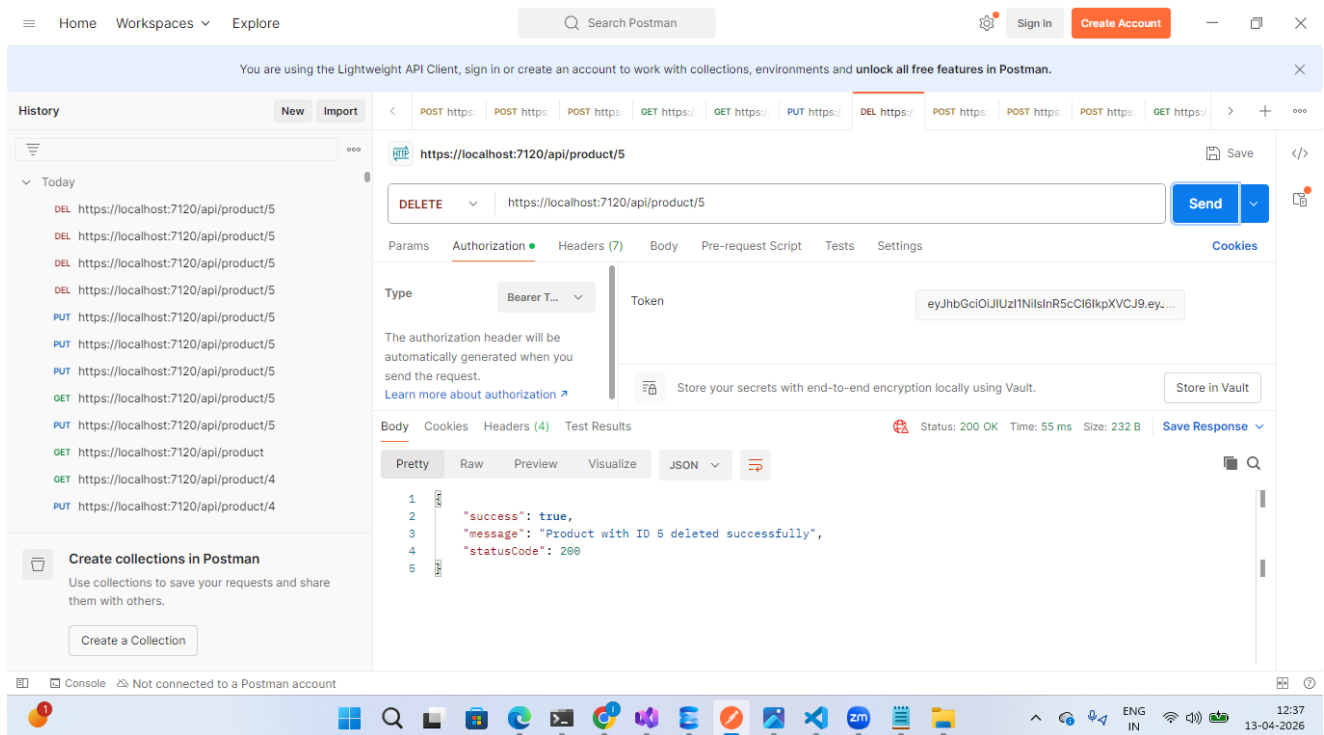


Screenshot 3.4h — Attempting to update a non-existent product ID returns 404 Not Found.

3.5 Delete Product (Admin Only)

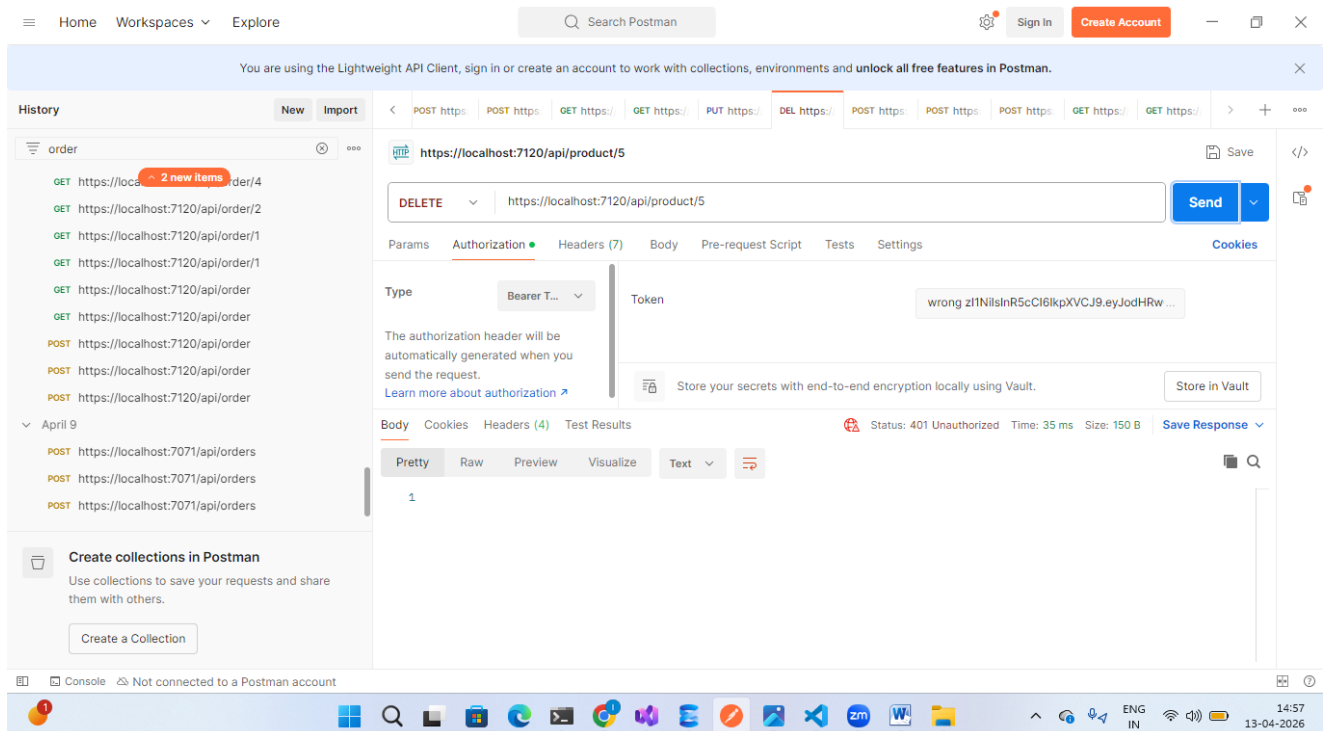
Successful Deletion — Product ID 5

With a valid admin JWT token, the product is deleted and a confirmation message is returned.



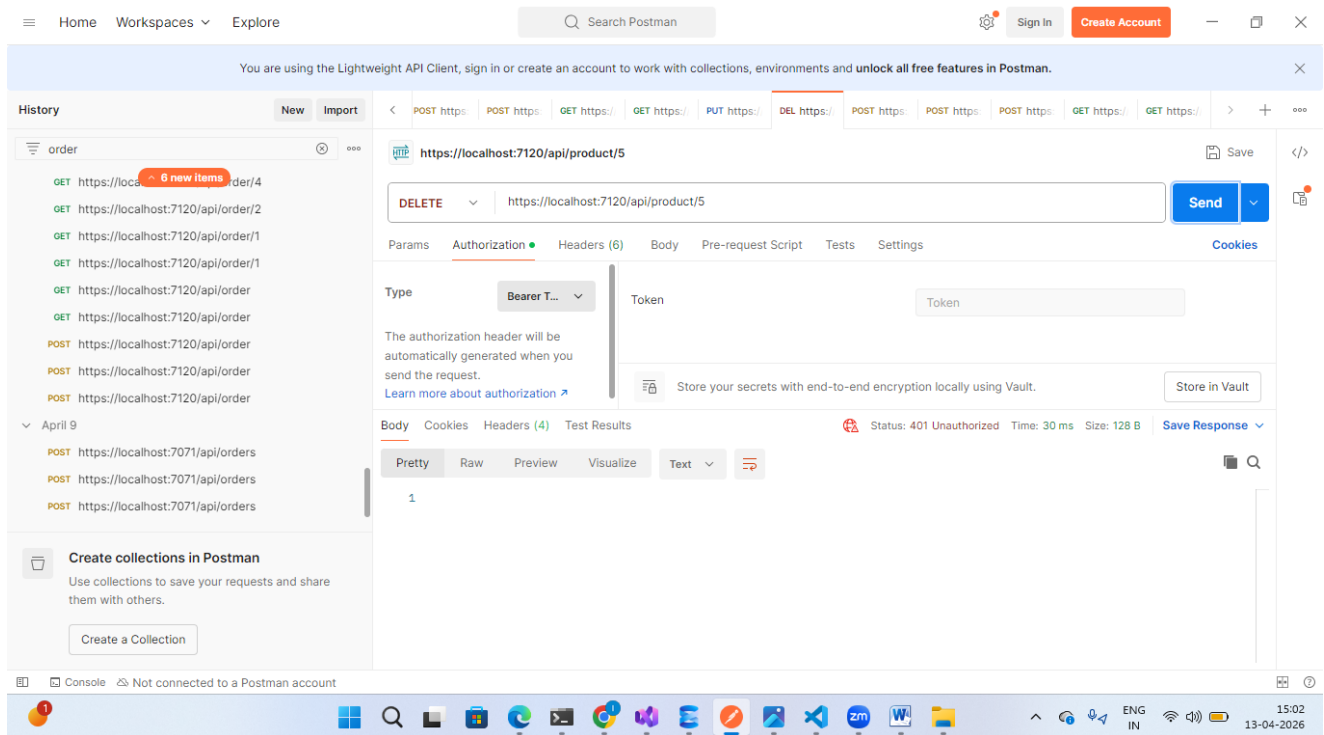
Screenshot 3.5a — Product ID 5 deleted successfully.

Wrong / User JWT Token



Screenshot 3.5b — User token or wrong token returns 403 Forbidden.

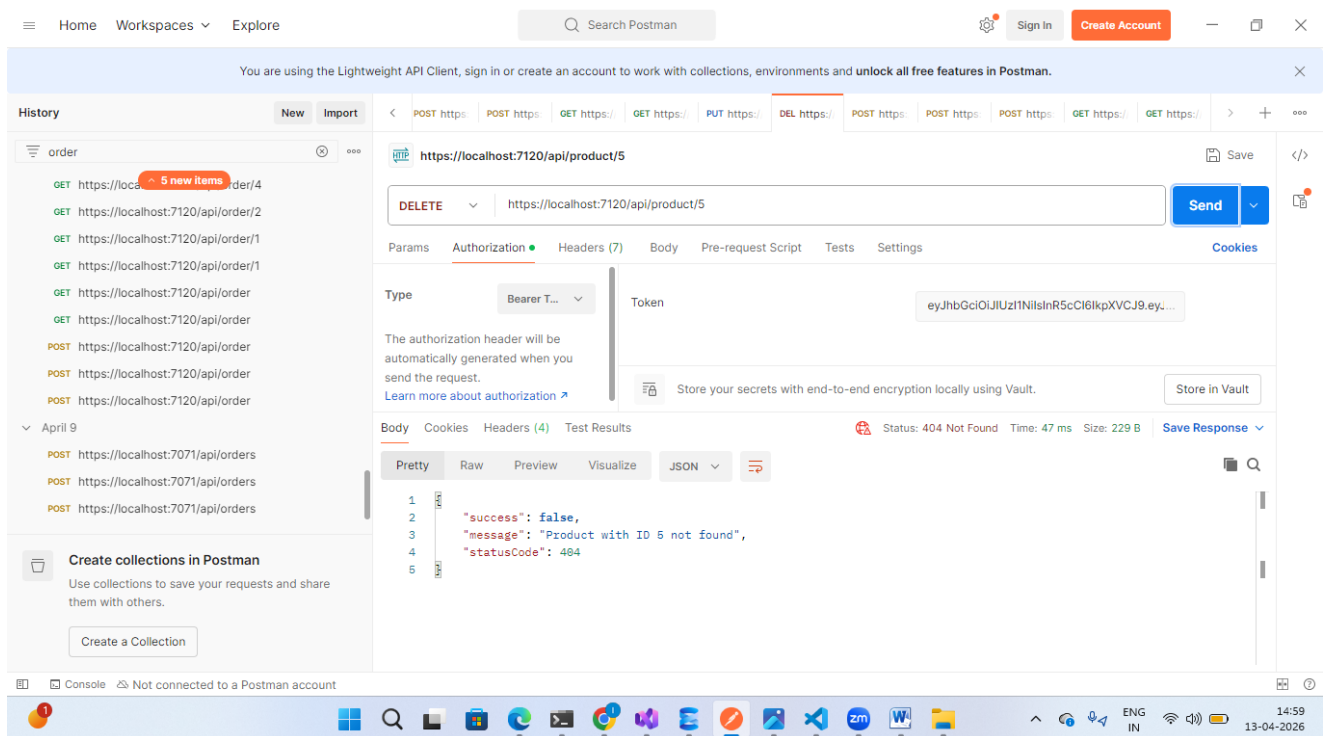
Missing Authorization



Screenshot 3.5c — Missing token returns 401 Unauthorized.

Verification — Product ID 5 No Longer Exists

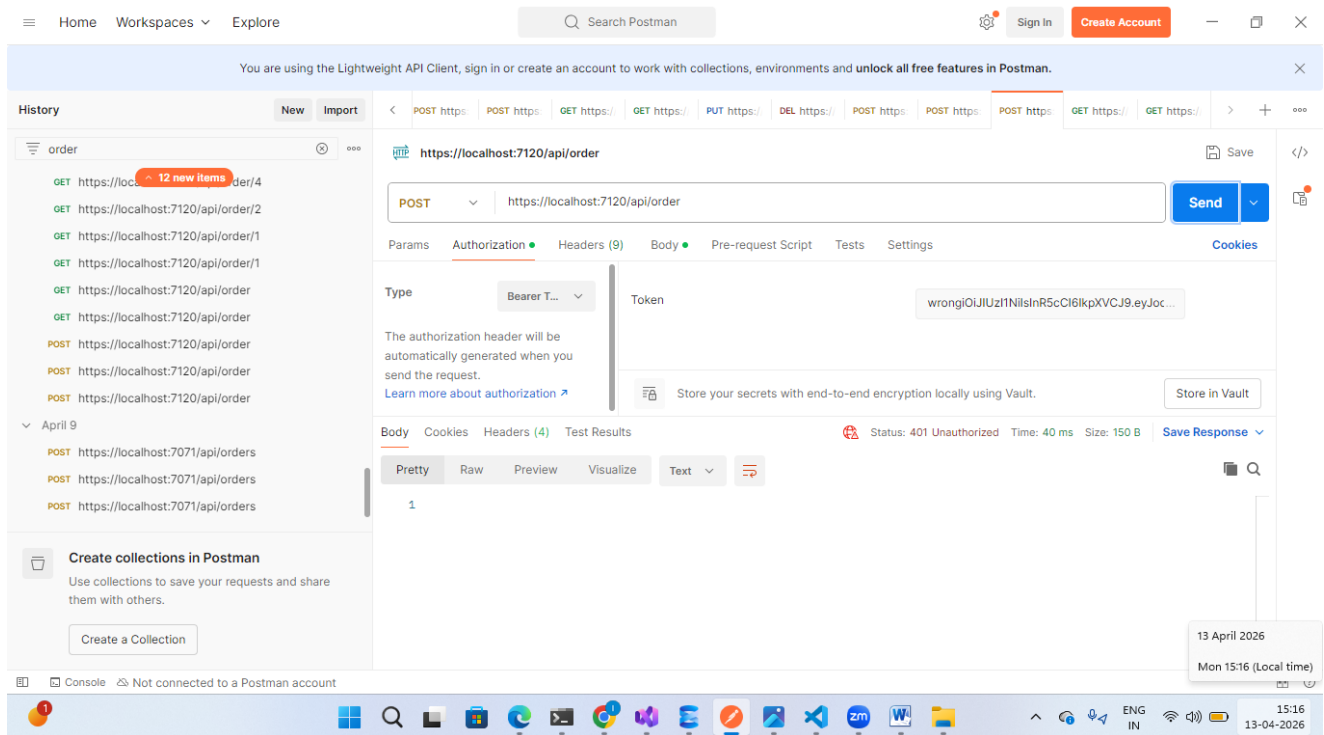
After deletion, a Get by ID request for product ID 5 confirms the record is removed.



Screenshot 3.5d — 404 Not Found confirms product ID 5 was deleted.

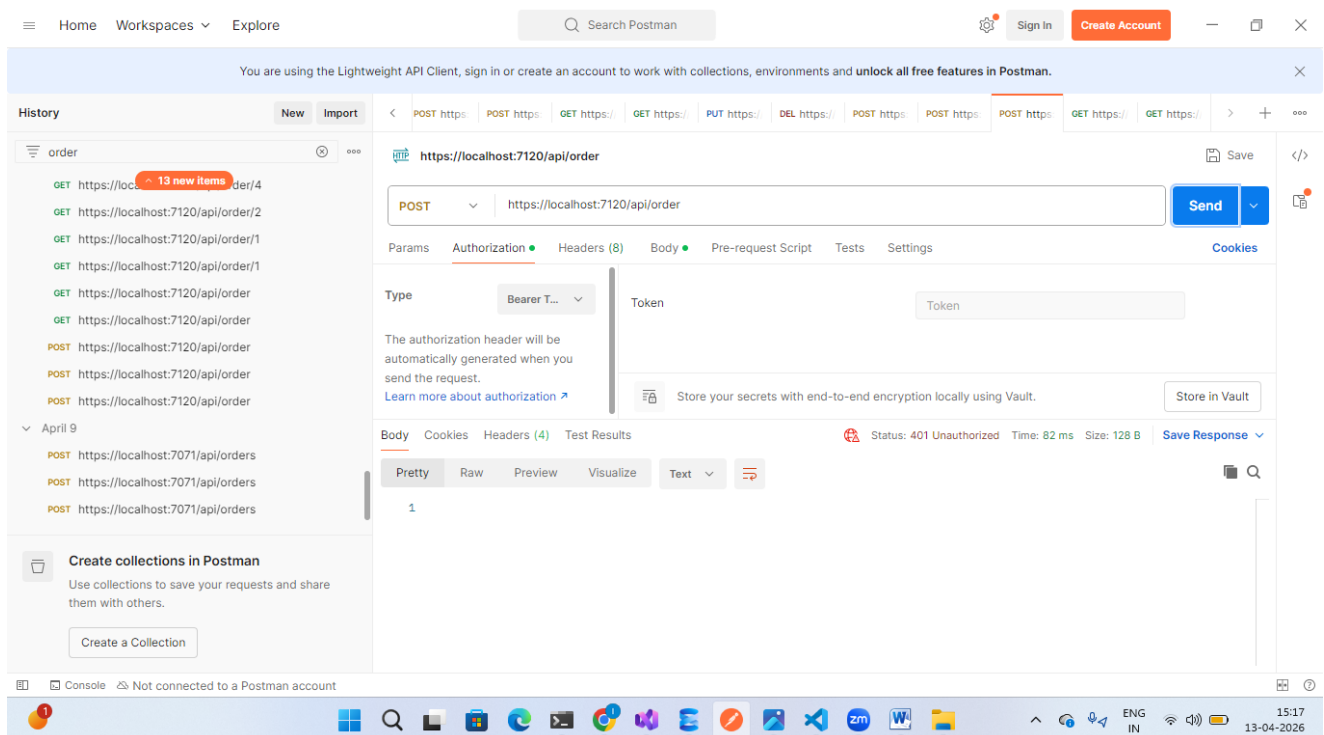
Delete Non-Existent Product (ID 91)

Invalid JWT Token



Screenshot 4.1b — Invalid JWT token returns 401 Unauthorized.

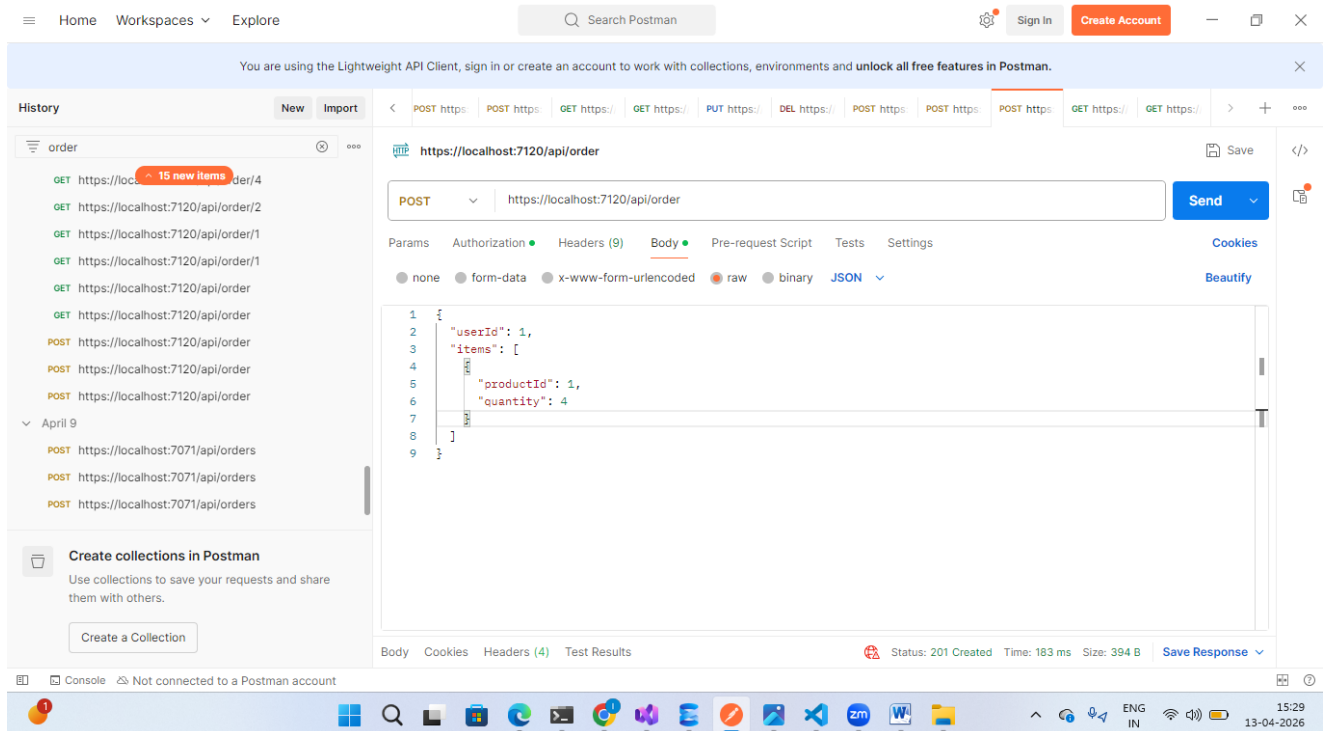
Missing Authorization



Screenshot 4.1c — Missing token returns 401 Unauthorized.

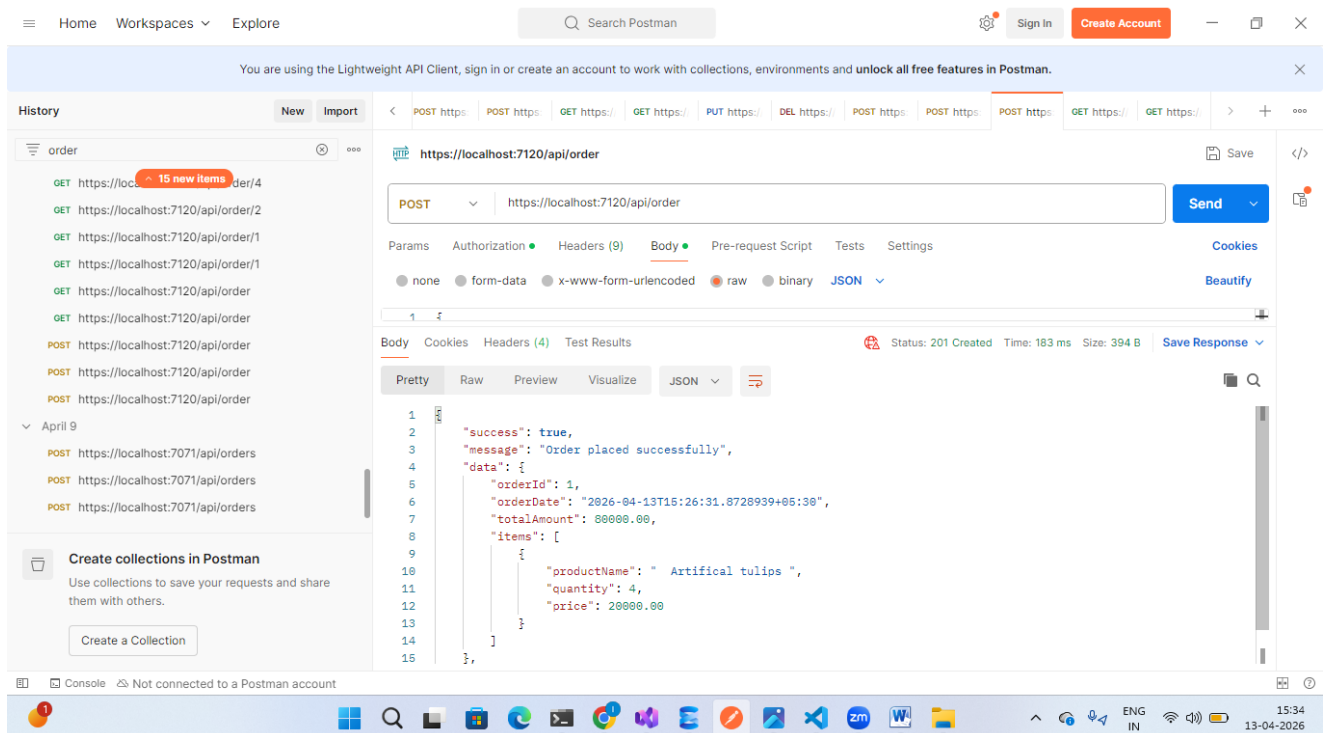
Order Request Body

The request body must include a valid product ID that exists in the database. The quantity must be a positive integer and must not exceed available stock.



Screenshot 4.1d — Order request body with valid product ID and quantity.

Order Placed Successfully



Screenshot 4.1e — Order placed successfully; stock updated automatically.

Invalid Product ID / Stock Exceeded

Ordering a non-existent product ID (e.g., 190) results in a 500 Internal Server Error. Requesting a quantity of 0 returns a validation error (quantity must be 1–100). If the requested quantity exceeds available stock (e.g., ordering 6 when only 5 remain), a descriptive error is returned. Stock is automatically decremented after each successful order.

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

order 17 new items

GET https://localhost:7120/api/order/4
GET https://localhost:7120/api/order/2
GET https://localhost:7120/api/order/1
GET https://localhost:7120/api/order/1
GET https://localhost:7120/api/order
GET https://localhost:7120/api/order
POST https://localhost:7120/api/order
POST https://localhost:7120/api/order
POST https://localhost:7120/api/order
April 9
POST https://localhost:7071/api/orders
POST https://localhost:7071/api/orders
POST https://localhost:7071/api/orders

Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection

https://localhost:7120/api/order

POST https://localhost:7120/api/order

Params Authorization Headers (9) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON Beautify

```
1 {
2   "userId": 1,
3   "items": [
4     {
5       "productId": 1,
6       "quantity": 0
7     }
8   ]
9 }
```

Body Cookies Headers (4) Test Results

Status: 400 Bad Request Time: 28 ms Size: 280 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "success": false,
3   "message": "Validation failed",
4   "errors": [
5     "The field Quantity must be between 1 and 100."
6   ],
7   "statusCode": 400
8 }
```

Console Not connected to a Postman account

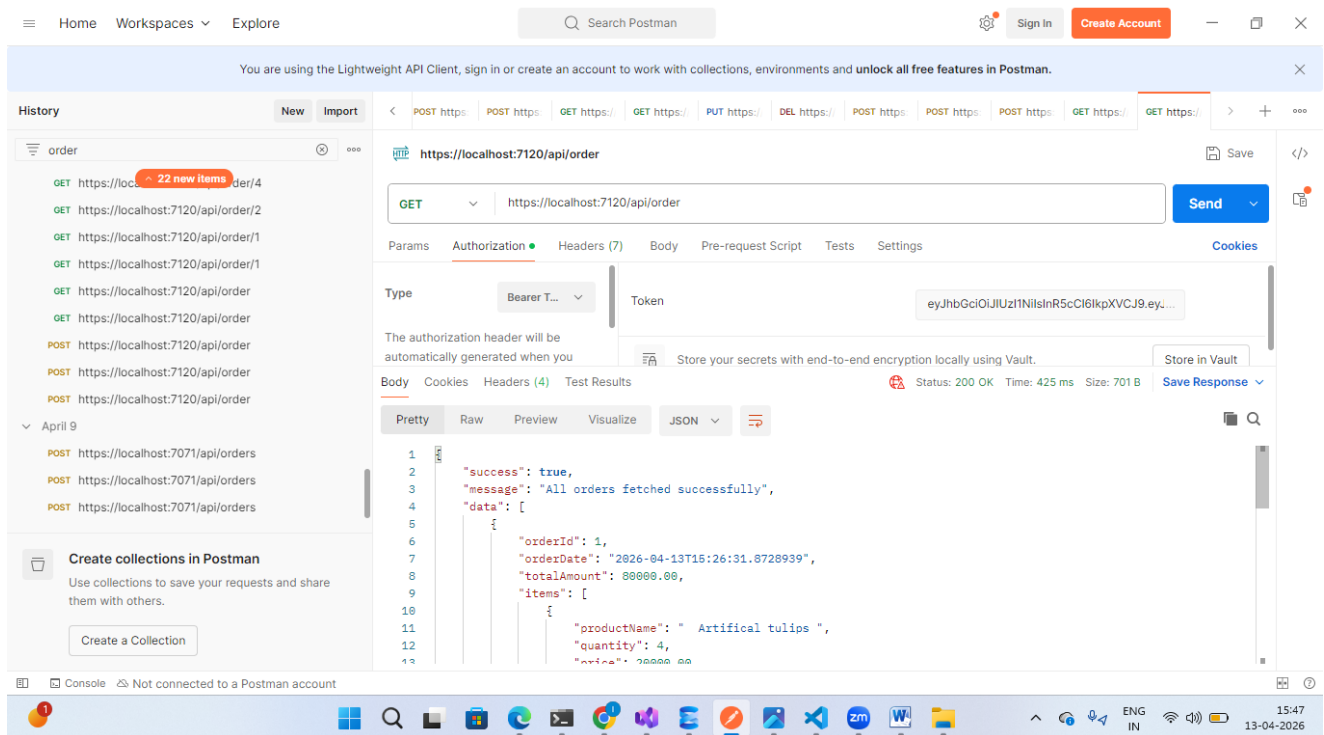
15:40 13-04-2026

Screenshot 4.1f — Error when placing order with invalid product ID or out-of-range quantity.

4.2 Get All Orders (Admin Only)

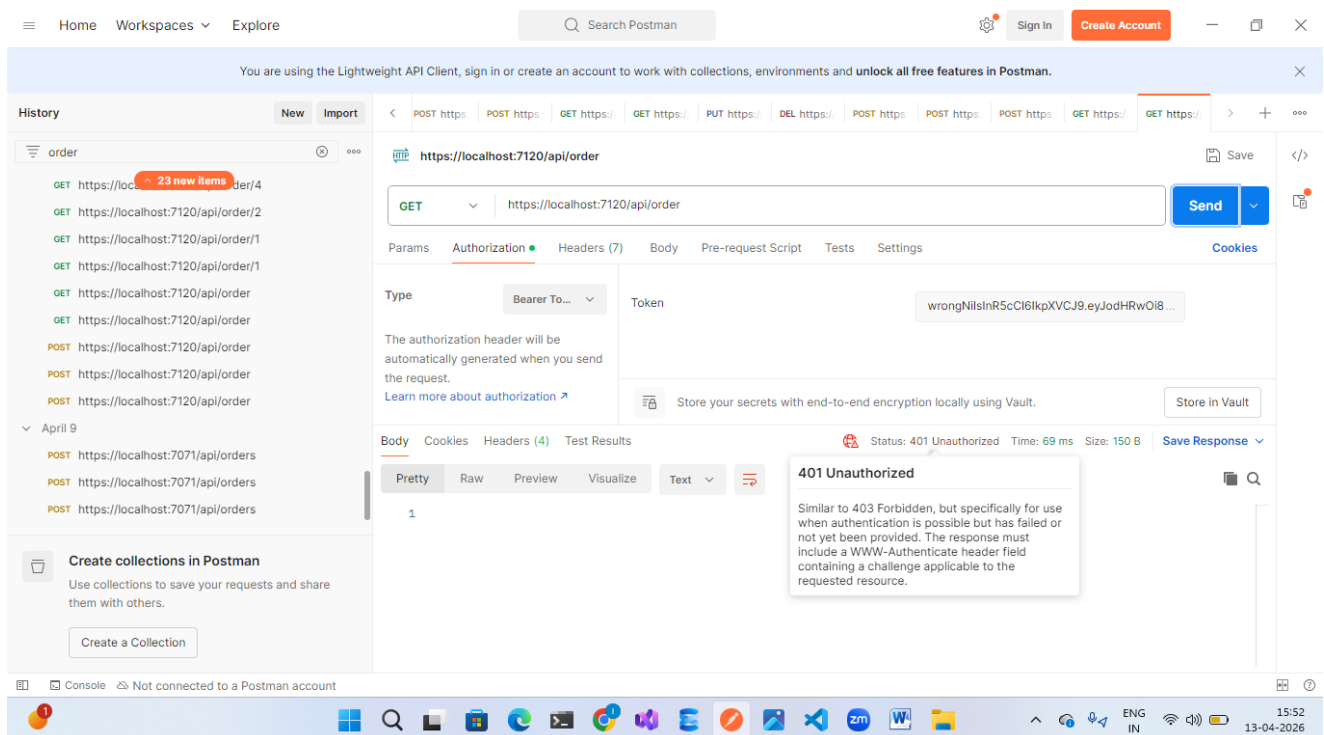
Only administrators can retrieve all orders. A valid admin JWT token is required.

Valid Admin Token



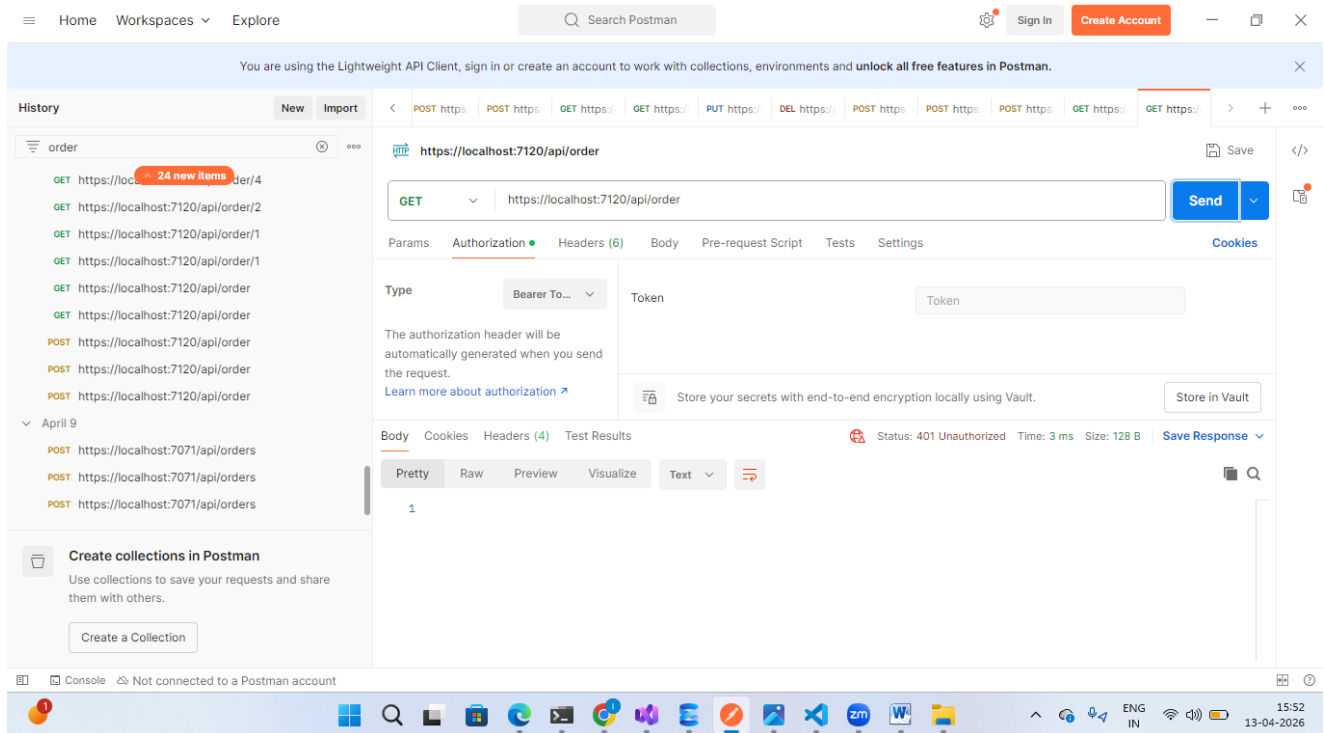
Screenshot 4.2a — All orders fetched with valid admin JWT token.

Wrong JWT Token



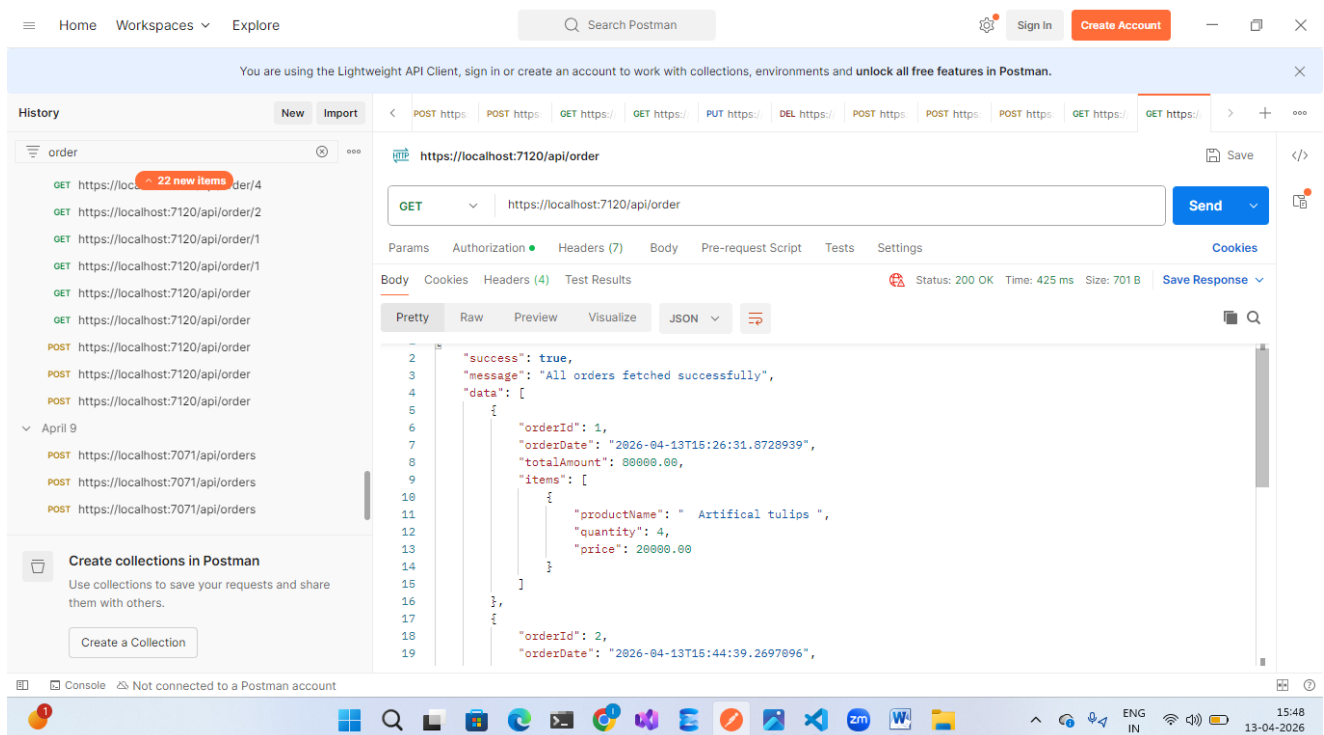
Screenshot 4.2b — Wrong JWT token returns 403 Forbidden.

Missing Authorization



Screenshot 4.2c — Missing token returns 401 Unauthorized.

Orders List Response



Screenshot 4.2d — Orders list (page 1).

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

order

GET https://localhost:7120/api/order/4
GET https://localhost:7120/api/order/2
GET https://localhost:7120/api/order/1
GET https://localhost:7120/api/order/1
GET https://localhost:7120/api/order
GET https://localhost:7120/api/order
POST https://localhost:7120/api/order
POST https://localhost:7120/api/order
POST https://localhost:7120/api/order
April 9
POST https://localhost:7071/api/orders
POST https://localhost:7071/api/orders
POST https://localhost:7071/api/orders

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https://localhost:7120/api/order

GET https://localhost:7120/api/order

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Body Cookies Headers (4) Test Results Status: 200 OK Time: 425 ms Size: 701 B Save Response

Pretty Raw Preview Visualize JSON

```
17 {
18   "orderId": 2,
19   "orderDate": "2026-04-13T15:44:39.2697896",
20   "totalAmount": 20000.00,
21   "items": [
22     {
23       "productName": "Artificial tulips ",
24       "quantity": 1,
25       "price": 20000.00
26     }
27   ]
28 },
29 {
30   "orderId": 3,
31   "orderDate": "2026-04-13T15:45:01.3608586",
32   "totalAmount": 799.00,
33   "items": [
34     {
35       "productName": "Modern Ceramic Vase",
36       "quantity": 1,
37       "price": 799.00
38     }
39   ]
40 },
41 ],
42 "statusCode": 200
43 }
```

Console Not connected to a Postman account

15:48 13-04-2026

Screenshot 4.2e — Orders list (page 2).

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

order

GET https://localhost:7120/api/order/4
GET https://localhost:7120/api/order/2
GET https://localhost:7120/api/order/1
GET https://localhost:7120/api/order/1
GET https://localhost:7120/api/order
GET https://localhost:7120/api/order
POST https://localhost:7120/api/order
POST https://localhost:7120/api/order
POST https://localhost:7120/api/order
April 9
POST https://localhost:7071/api/orders
POST https://localhost:7071/api/orders
POST https://localhost:7071/api/orders

Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection

https://localhost:7120/api/order

GET https://localhost:7120/api/order

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Body Cookies Headers (4) Test Results Status: 200 OK Time: 425 ms Size: 701 B Save Response

Pretty Raw Preview Visualize JSON

```
20   "price": 20000.00
26   }
27   ]
28   },
29   {
30     "orderId": 3,
31     "orderDate": "2026-04-13T15:45:01.3608586",
32     "totalAmount": 799.00,
33     "items": [
34       {
35         "productName": "Modern Ceramic Vase",
36         "quantity": 1,
37         "price": 799.00
38       }
39     ]
40   },
41 ],
42 "statusCode": 200
43 }
```

Console Not connected to a Postman account

15:48 13-04-2026

Screenshot 4.2f — Orders list (page 3).

4.3 Get Order by ID

Both admin and user tokens can retrieve a specific order by its ID.

The screenshot shows the Postman interface with a GET request to `https://localhost:7120/api/order/2`. The request is authenticated with a Bearer token. The response is a 200 OK status with a JSON body containing order details.

```
2  {
3    "success": true,
4    "message": "Order retrieved successfully",
5    "data": {
6      "orderId": 2,
7      "orderDate": "2026-04-13T15:44:39.2697096",
8      "totalAmount": 20000.00,
9      "items": [
10       {
11         "productName": "Artificial tulips ",
12         "quantity": 1,
13         "price": 20000.00
14       }
15     ]
16   },
17   "statusCode": 200
18 }
```

Screenshot 4.3a — Order ID 2 retrieved successfully.

Non-Existent Order ID

The screenshot shows the Postman interface with a GET request to `https://localhost:7120/api/order/22`. The request is authenticated with a Bearer token. The response is a 404 Not Found status with a JSON body indicating the order was not found.

```
1  {
2    "success": false,
3    "message": "Order with ID 22 not found",
4    "statusCode": 404
5  }
```

Screenshot 4.3b — 404 Not Found for an order ID that does not exist.

5. Advanced Features — Pagination, Search & Filtering

Authorization for Product APIs

Why Authorization is NOT Required for Pagination, Search, and Filtering

The following operations — Pagination, Search (by product name), and Filtering (by price) — are all implemented under the **GET /api/product** endpoint. These operations do not require authentication or authorization because they are **read-only operations**.

Reasoning

In real-world e-commerce applications, users can browse products without logging in. Searching and filtering are integral parts of the browsing experience, and these operations do not modify any data. Therefore, exposing these APIs publicly improves user experience and aligns with industry standards.

Technical Implementation

This is achieved by applying **[AllowAnonymous]** on the GET endpoint inside **ProductController**. This attribute overrides the controller-level **[Authorize]** and allows public access to the endpoint without any token.

Security Consideration

While read operations are publicly accessible, all write operations remain protected. Create Product, Update Product, and Delete Product are restricted to administrators using the following attribute:

Summary

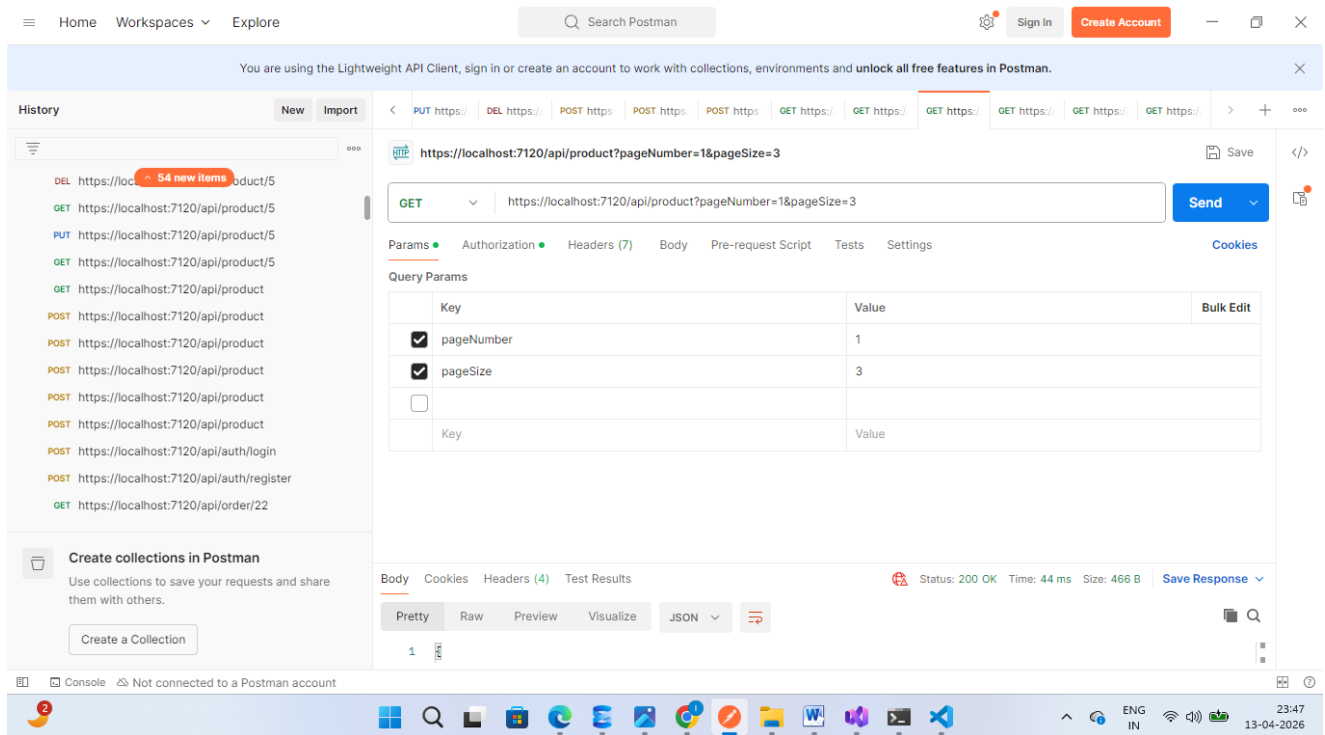
Conclusion

Pagination, search, and filtering are public APIs because they are non-sensitive, read-only operations that enhance usability without compromising security. This design pattern is consistent with how major e-commerce platforms such as Amazon, Flipkart, and Myntra operate — product browsing is always available without requiring a login.

5.1 Pagination

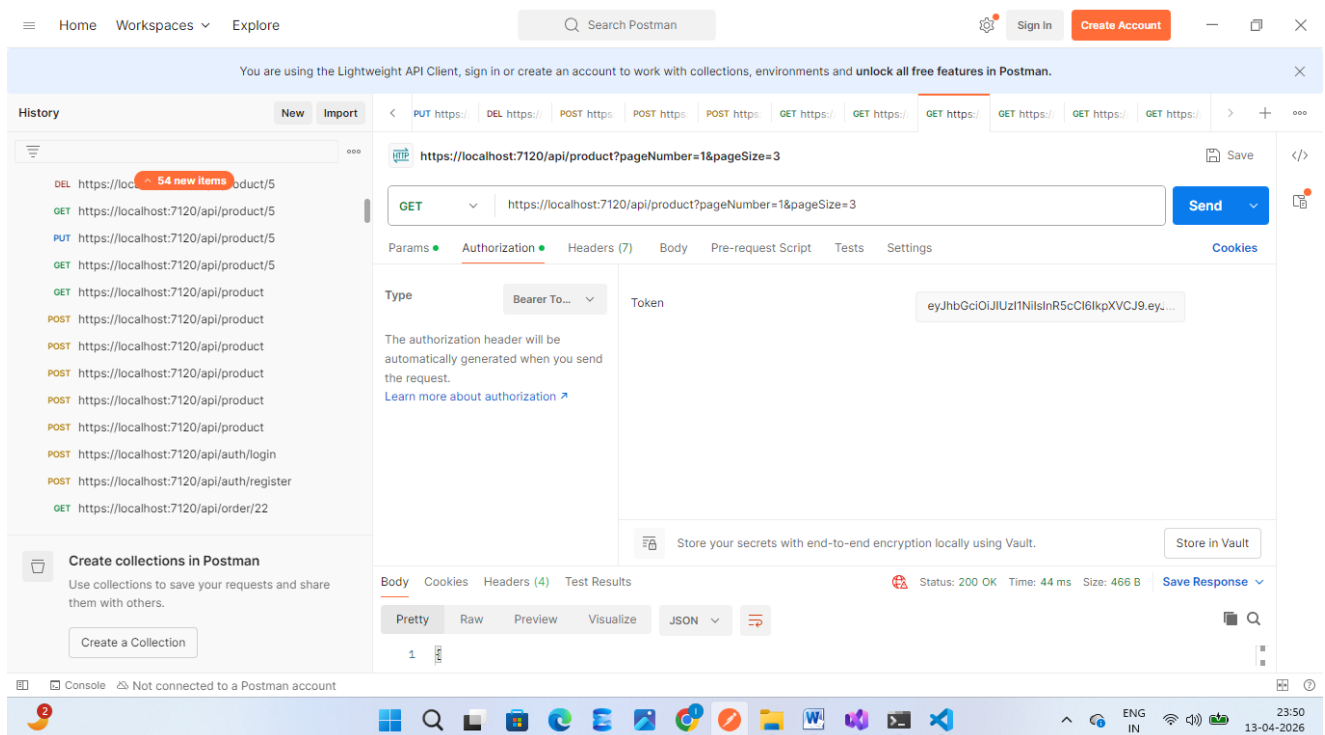
Pagination limits the number of records per response. Use the 'pageNumber' and 'pageSize' query parameters.

Params



Screenshot 5.1a — Pagination query parameters: pageNumber and pageSize.

Authorization (Optional)



Screenshot 5.1b — Request with valid JWT token.

No Token — Still Works

HomeWorkspacesExplore

Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

HistoryNewImport

56 new items

DEL https://localhost:7120/api/product/5

GET https://localhost:7120/api/product/5

PUT https://localhost:7120/api/product/5

GET https://localhost:7120/api/product/5

GET https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/auth/login

POST https://localhost:7120/api/auth/register

GET https://localhost:7120/api/order/22

Create collections in Postman

https://localhost:7120/api/product?pageNumber=1&pageSize=3

GEThttps://localhost:7120/api/product?pageNumber=1&pageSize=3

Send

ParamsAuthorizationHeaders (6)BodyPre-request ScriptTestsSettingsCookies

Bearer Token

The authorization header will be automatically generated when you send the request.

Store your secrets with end-to-end encryption locally using Vault.

Store in Vault

BodyCookiesHeaders (4)Test Results

Status: 200 OKTime: 17 msSize: 785 BSave Response

PrettyRawPreviewVisualizeJSON

```
1  {
2    "success": true,
3    "message": "Products fetched successfully",
4    "data": {
5      "totalRecords": 4,
6      "pageNumber": 1,
7      "pageSize": 3,
8      "data": [
9        {
10         "productId": 1,
11         "name": "Premium Sofa Set"
```

ConsoleNot connected to a Postman account

23:5313-04-2026

Screenshot 5.1c — Request without token succeeds due to [AllowAnonymous] on the GET endpoint.

5.2 Search by Product Name

The 'search' parameter allows partial-match searching by product name, enabling users to find relevant products quickly.

Authorization

The screenshot shows the Postman interface with a GET request to `https://localhost:7120/api/product?search=sofa`. The request is configured with a Bearer token authorization header. The response is displayed in the 'Pretty' tab, showing a JSON object with a list of products.

Request:

```
GET https://localhost:7120/api/product?search=sofa
```

Authorization: Bearer Token: `eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1b250IjoiIiwiaWF0IjoiMTY1MjM0MjM0In0.eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1b250IjoiIiwiaWF0IjoiMTY1MjM0MjM0In0.`

Response (Pretty):

```
{
  "data": {
    "totalRecords": 1,
    "pageNumber": 1,
    "pageSize": 5,
    "data": [
      {
        "productId": 1,
        "name": "Premium Sofa Set",
        "description": "Comfortable fabric sofa set with wooden frame and modern design.",
        "price": 32000.00,
      }
    ]
  }
}
```

Screenshot 5.2a — Authorization header for search request.

Params

The screenshot shows the Postman interface with a GET request to `https://localhost:7120/api/product?search=sofa`. The request is configured with a query parameter 'search' set to 'sofa'. The response is displayed in the 'Pretty' tab, showing a JSON object with a list of products.

Request:

```
GET https://localhost:7120/api/product?search=sofa
```

Query Params:

Key	Value
search	sofa

Response (Pretty):

```
{
  "data": {
    "totalRecords": 1,
    "pageNumber": 1,
    "pageSize": 5,
    "data": [
      {
        "productId": 1,
        "name": "Premium Sofa Set",
        "description": "Comfortable fabric sofa set with wooden frame and modern design.",
        "price": 32000.00,
      }
    ]
  }
}
```

Screenshot 5.2b — Search query parameter.

Search Response

HomeWorkspacesExplore

Search Postman

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You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

61 new items

product/5

GET https://localhost:7120/api/product/5

PUT https://localhost:7120/api/product/5

GET https://localhost:7120/api/product/5

GET https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/api/product

POST https://localhost:7120/auth/login

POST https://localhost:7120/auth/register

GET https://localhost:7120/api/order/22

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

https://localhost:7120/api/product?search=sofa

GET

Send

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettings

BodyCookiesHeaders (4)Test Results

Status: 200 OKTime: 11 msSize: 466 BSave Response

PrettyRawPreviewVisualizeJSON

```
2  "success": true,
3  "message": "Products fetched successfully",
4  "data": {
5    "totalRecords": 1,
6    "pageNumber": 1,
7    "pageSize": 5,
8    "data": [
9      {
10       "productId": 1,
11       "name": "Premium Sofa Set",
12       "description": "Comfortable fabric sofa set with wooden frame and modern design.",
13       "price": 32000.00,
14       "imageUrl": "images/sofa.jpg",
15       "stock": 12
16     }
17   ]
18 }
19 "statusCode": 200
```

Console

Not connected to a Postman account

00:0014-04-2026

Screenshot 5.2c — Products matching the search term returned successfully.

5.3 Filter by Price Range

Products can be filtered by minimum and/or maximum price using 'minPrice' and 'maxPrice' query parameters.

Params

The screenshot shows the Postman interface with a GET request to `https://localhost:7120/api/product?minPrice=1000&maxPrice=400000`. The 'Query Params' tab is active, displaying a table with the following data:

Key	Value
☒ minPrice	1000
☒ maxPrice	400000
Key	Value

The 'Body' tab is also visible, showing the response status: 200 OK, Time: 86 ms, Size: 968 B. The response is displayed in 'Pretty' format.

Screenshot 5.3a — Price range filter parameters.

Authorization

The screenshot shows the same GET request in Postman, but with the 'Authorization' tab selected. The 'Type' is set to 'Bearer Token' and the 'Token' field contains the value `eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1b250Ijoi...`. The 'Body' tab is also visible, showing the response status: 200 OK, Time: 86 ms, Size: 968 B. The response is displayed in 'Pretty' format.

Screenshot 5.3b — Authorization header for price filter request.

Filter Response

Home Workspaces Explore Search Postman

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History: **61 new items**

- DEL https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product/5
- PUT https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/register
- GET https://localhost:7120/api/order/22

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

URL: `https://localhost:7120/api/product?minPrice=1000&maxPrice=400000`

Method: GET

Send

Params: Authorization: Headers (7) Body Pre-request Script Tests Settings Cookies

Status: 200 OK Time: 86 ms Size: 968 B Save Response

Body: Pretty Raw Preview Visualize JSON

```
1  {
2    "success": true,
3    "message": "Products fetched successfully",
4    "data": {
5      "totalRecords": 4,
6      "pageNumber": 1,
7      "pageSize": 5,
8      "data": [
9        {
10         "productId": 1,
11         "name": "Premium Sofa Set",
12         "description": "Comfortable fabric sofa set with wooden frame and modern design.",
13         "price": 32000.00,
14         "imageUrl": "images/sofa.jpg",
15         "stock": 12
16       },
17       {
18         "productId": 2,
```

Screenshot 5.3c — Filtered products (result page 1).

Home Workspaces Explore Search Postman

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History: **61 new items**

- DEL https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product/5
- PUT https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/register
- GET https://localhost:7120/api/order/22

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

URL: `https://localhost:7120/api/product?minPrice=1000&maxPrice=400000`

Method: GET

Send

Params: Authorization: Headers (7) Body Pre-request Script Tests Settings Cookies

Status: 200 OK Time: 86 ms Size: 968 B Save Response

Body: Pretty Raw Preview Visualize JSON

```
17  {
18    "productId": 2,
19    "name": "Decorative Table Lamp",
20    "description": "Elegant lamp for bedroom decor",
21    "price": 1899.00,
22    "imageUrl": "lamp.jpg",
23    "stock": 1
24  },
25  {
26    "productId": 3,
27    "name": "LED Bathroom Mirror",
28    "description": "Round LED mirror with anti-fog feature and touch controls.",
29    "price": 5999.00,
30    "imageUrl": "images/mirror.jpg",
31    "stock": 6
32  },
33  {
34    "productId": 4,
```

Screenshot 5.3d — Filtered products (result page 2).

Home Workspaces Explore Search Postman

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

- DEL https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product/5
- PUT https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product/5
- GET https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/product
- POST https://localhost:7120/api/auth/login
- POST https://localhost:7120/api/auth/register
- GET https://localhost:7120/api/order/22

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

GET https://localhost:7120/api/product?minPrice=1000&maxPrice=400000

Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Body Cookies Headers (4) Test Results

Status: 200 OK Time: 86 ms Size: 968 B Save Response

Pretty Raw Preview Visualize JSON

```
26
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[
  {
    "productId": 3,
    "name": "LED Bathroom Mirror",
    "description": "Round LED mirror with anti-fog feature and touch controls.",
    "price": 5999.00,
    "imageUrl": "images/mirror.jpg",
    "stock": 6
  },
  {
    "productId": 4,
    "name": "Wall Mounted Wooden Shelf",
    "description": "Minimal floating shelf perfect for displaying decor items.",
    "price": 2499.00,
    "imageUrl": "images/shelf.jpg",
    "stock": 12
  }
],
{
  "statusCode": 200
}
```

Console Not connected to a Postman account

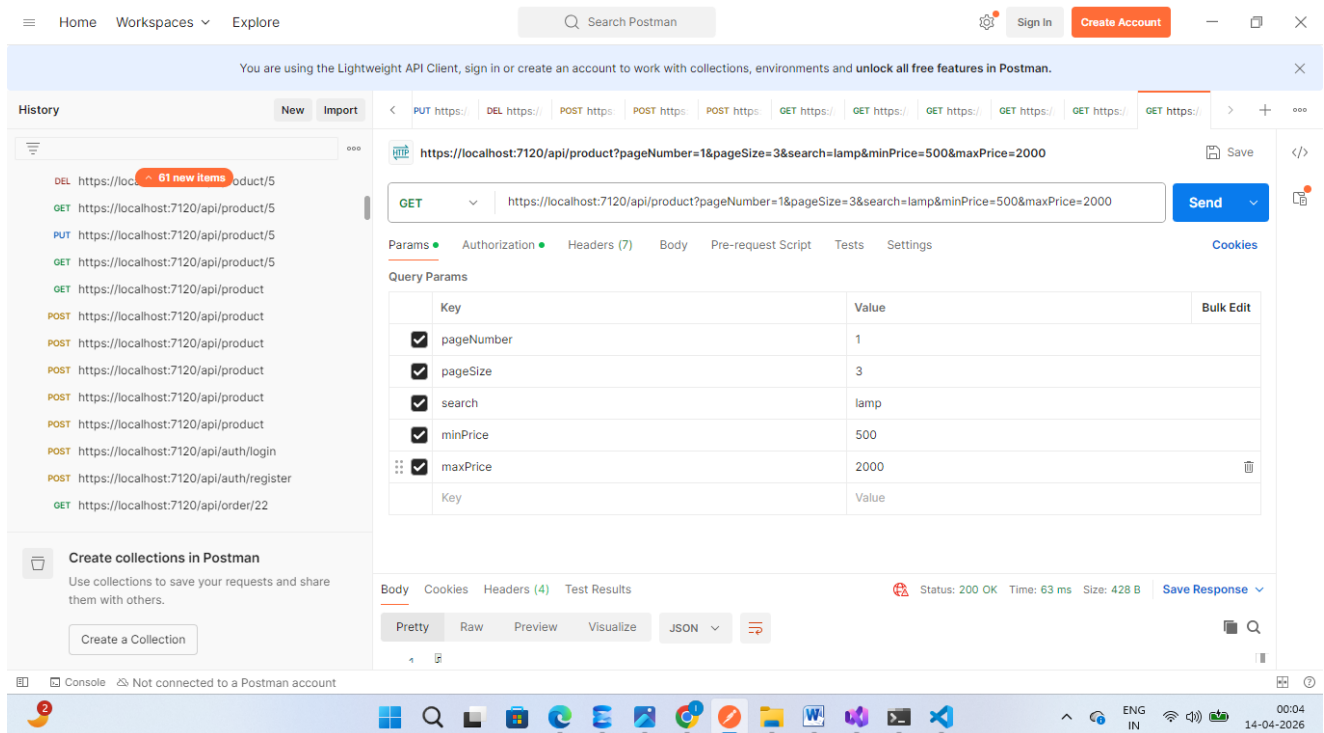
00:03 14-04-2026

Screenshot 5.3e — Filtered products (result page 3).

5.4 Combined: Pagination + Search + Filtering

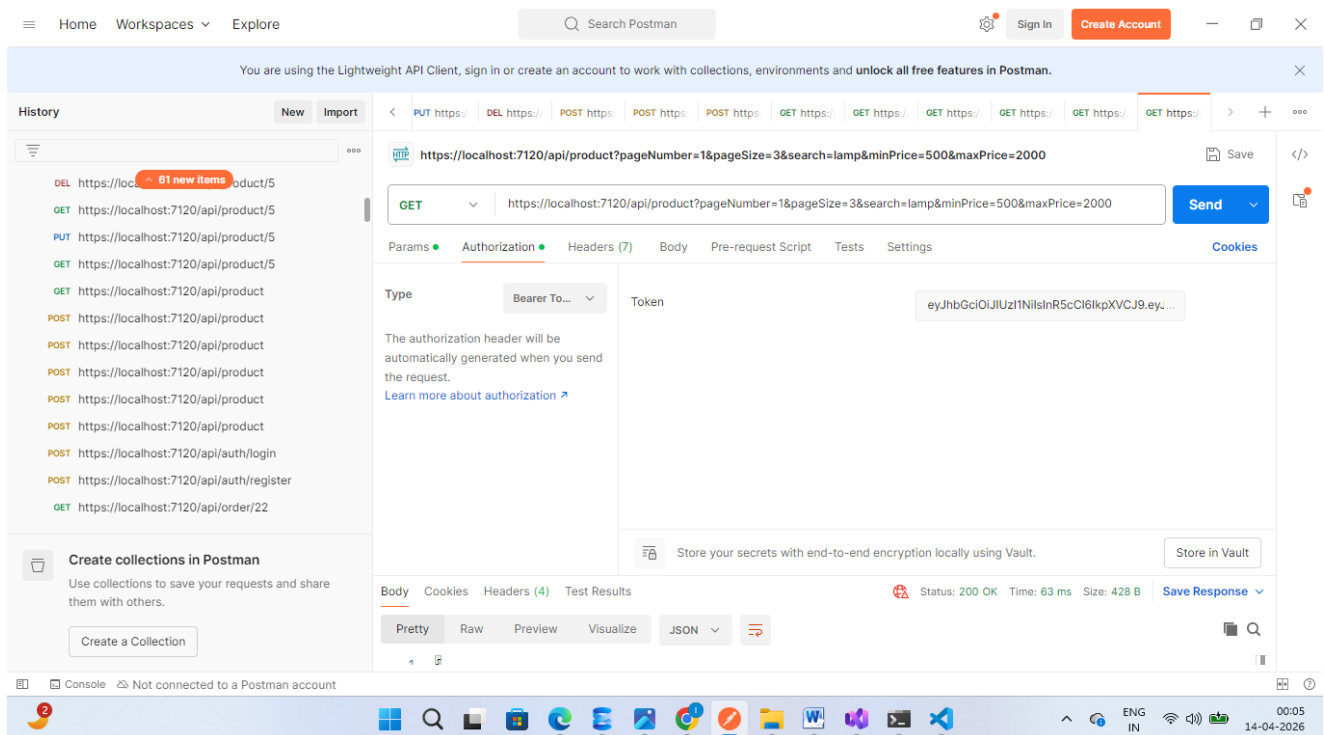
All three features can be combined in a single request. Since they all map to the same `GetProducts(ProductQueryDTO query)` method, any combination of parameters is supported.

Params



Screenshot 5.4a — Combined query parameters for pagination, search, and price filter.

Authorization



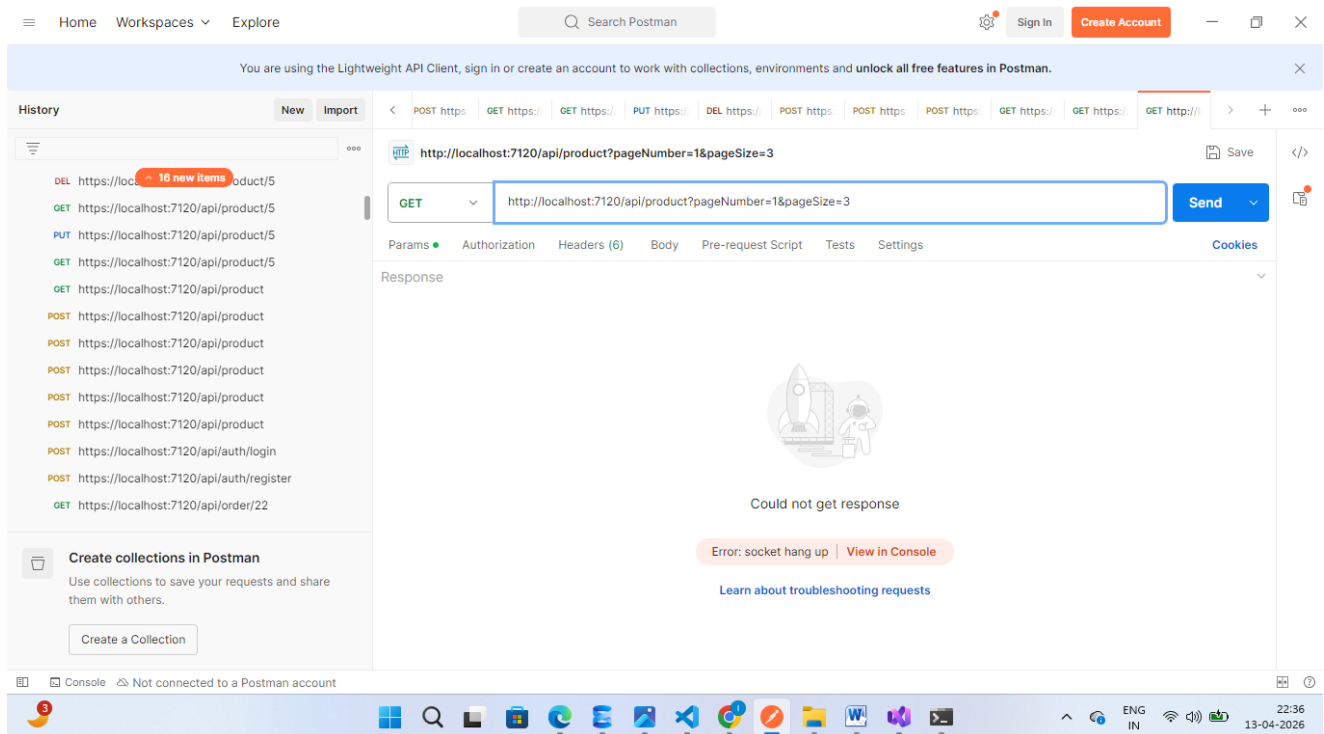
Screenshot 5.4b — Authorization header.

Screenshot 5.4c — Correct results returned for combined query.

6. Common Error Reference & Troubleshooting

6.1 ERR_CONNECTION_REFUSED / Unable to Connect

This is the most frequent error encountered when the API server is not running. The screenshot below shows this error in Postman.



Screenshot 6.1 — Connection refused error when API is not running.

Troubleshooting Steps

Step 1 — Check Visual Studio: Confirm the project is actively running (green play button active). If not, click Run (■).

Step 2 — Check Output / Terminal: Look for exception or crash logs. If the API has crashed, resolve the error in code before retesting.

Step 3 — Verify Port: Ensure the port in Postman matches the port in Visual Studio (e.g., `http://localhost:7120`). Test the base URL directly in a browser first.

Step 4 — HTTP vs HTTPS: Try both `http://` and `https://` variants if one does not respond.

Step 5 — Simplify the Request: Strip all query parameters and test the bare endpoint (`GET /api/product`) before adding pagination, search, or filter params.

6.2 Quick Error Code Summary

HTTP Status	Meaning	Typical Cause
200 OK	Success	Request processed correctly
201 Created	Resource Created	POST/registration succeeded
400 Bad Request	Validation Error	Missing or invalid input fields

401 Unauthorized	Authentication Failed	Missing or invalid JWT token
403 Forbidden	Authorization Failed	Valid token but insufficient role (e.g., user using admin endpoint)
404 Not Found	Resource Missing	Product/Order ID does not exist
500 Internal Error	Server Crash	Unhandled exception (e.g., invalid foreign key)

6.3 Authorization Rules Summary

Feature / Endpoint	Admin Token	User Token	No Token
Register (Admin/User)	—	—	Public
Login (Admin/User)	—	—	Public
Create Product	✓	✗	✗
Update Product	✓	✗	✗
Delete Product	✓	✗	✗
View All Products	✓	✓	✓ (AllowAnonymous)
Search Products	✓	✓	✓ (AllowAnonymous)
Filter by Price	✓	✓	✓ (AllowAnonymous)
Pagination	✓	✓	✓ (AllowAnonymous)
Place Order	✗	✓	✗
Get All Orders	✓	✗	✗
Get Order by ID	✓	✓	✗